

Testing Cointegration with Many Persistent Controls

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Abstract

This paper proposes DML-Bounds, a procedure for testing cointegration in data-rich time-series settings with many persistent controls. The autoregressive distributed lag (ARDL) bounds test is attractive because it avoids pretesting the integration order of the regressors, but it is not designed for a high-dimensional conditioning set. We show that residualizing lagged levels against persistent controls can absorb stochastic trends and thereby change the finite-sample null distribution of the bounds statistic. The object governing inference is then not the integration order of the original regressors but the effective number of stochastic trends that survive residualization, which we formalize as an effective integrated count $\tilde{k}$. This yields a trend-absorption bracket: stationary controls leave the classical upper-bound logic intact, while integrated controls can move the relevant null inside the Pesaran–Shin–Smith interval. The procedure combines h -block cross-fitting, a balanced nuisance projection, and a restricted *system* wild bootstrap that regenerates the dependent variable and the focal regressor jointly, with a common wild weight applied to the conditional and marginal residuals, and recomputes the full residualized statistic under the null. Joint regeneration preserves the contemporaneous correlation between the marginal innovations of the regressor and the conditional error, so the bootstrap remains valid when the focal regressor is endogenous, a central feature of the Pesaran–Shin–Smith setting that a fixed-regressor scheme cannot replicate; consistency of the system bootstrap is established in the oracle fixed-integrated-block design for any value of the endogeneity correlation. Monte Carlo experiments show that classical bounds critical values can severely over-reject when the nuisance controls are integrated, whereas the bootstrap restores size and retains useful power, including under strong endogeneity of the focal regressor. The main application is exchange-rate pass-through to U.S. prices across four monetary regimes, where the control-set diagnostic flips the verdict in the late Great Moderation: dropping the controls that share trends with the dollar turns a non-rejection into a rejection, showing how high-dimensional nuisance adjustment changes cointegration inference in practice. A canonical consumption–income illustration is reported as a supporting check.

Keywords: Cointegration; High-dimensional econometrics; Double machine learning; Orthogonalization; ARDL; Bounds testing; LASSO; Wild bootstrap; Exchange-rate pass-through.

JEL codes: C22; C13; C55; C58; F31; E31.

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1 Introduction

Understanding long-run equilibrium relationships between economic variables lies at the core of empirical macroeconomics and finance. Cointegration analysis, formalized by [Engle and Granger \(1987\)](#), provides a framework for studying such relationships among non-stationary variables. In its classical formulation, cointegration arises when a linear combination of integrated processes eliminates common stochastic trends and yields a stationary residual, thereby representing a stable long-run equilibrium.

In many empirical applications, however, the variables of interest are embedded in a much richer information set. Researchers often observe a large number of covariates capturing macroeconomic conditions, sectoral dynamics, financial factors, and other sources of persistent common variation. These covariates may include both stationary $I(0)$ and non-stationary $I(1)$ components. When such confounding variation is omitted or handled inadequately, conventional cointegration tests may either fail to detect genuine long-run relationships or spuriously suggest their presence.

Traditional approaches, including the autoregressive distributed lag (ARDL) bounds-testing framework of [Pesaran et al. \(2001\)](#), are not designed for environments in which the relevant nuisance space is high-dimensional, so that conditioning on it raises the classic difficulties of estimation in high dimensions ([Bellman, 1961](#)). While ARDL is attractive because it accommodates mixtures of $I(0)$ and $I(1)$ variables and models short-run dynamics explicitly, its classical unpenalized implementation becomes fragile when the conditioning set is large relative to the sample size. In such settings, the projection step required for long-run inference may become unstable or mathematically infeasible.

At the same time, advances in high-dimensional econometrics, particularly the Double Machine Learning (DML) framework of [Chernozhukov et al. \(2018\)](#), provide tools for inference in the presence of complex nuisance functions. These methods rely on orthogonalization and sample splitting to reduce sensitivity to first-stage regularization bias. Although DML has been developed primarily for cross-sectional and stationary settings, its projection logic offers a tractable route for extending cointegration analysis to richer time-series environments. The key complication, and the focus of this paper, is that in an integrated environment the projection step is not innocuous: it can change the very object that governs the test.

Contribution. This paper contributes to the applied cointegration-testing literature by showing that, in data-rich time-series settings, the finite-sample null distribution of an ARDL-style bounds statistic can be materially altered by residualizing lagged levels against persistent controls. The proposed DML-Bounds procedure combines balanced high-dimensional residualization with a restricted system wild bootstrap, providing a feasible testing strategy when conventional ARDL conditioning sets are too small and unpenalized high-dimensional error-correction models are unstable or infeasible. The name is shorthand for the orthogonalization and cross-fitting logic the procedure borrows; what the paper develops is orthogonalized bounds testing with high-dimensional persistent controls, not a standard double machine learning problem with a low-dimensional orthogonal score, a distinction Section 4 makes precise. At the level of theory, the paper develops a bounds-testing account of cointegration after high-dimensional orthogonalization. Classical ARDL bounds tests bracket uncertainty about whether the tested regressors are stationary or integrated. In high-dimensional empirical settings a second source of uncertainty appears: nuisance controls may themselves contain stochastic trends, and partialling out those controls can absorb part of the long-run variation that identifies the error-correction relation. The object that governs inference is therefore no longer only the integration order of the original regressors, but the effective number of stochastic trends that survive residualization. The paper formalizes this object as an effective integrated count $\tilde{k}$ and shows that the orthogonalized bounds statistic converges to a bounds-type Brownian functional whose stochastic-trend dimension is $\tilde{k}$, so that residualization shifts the relevant null reference from the classical k -integrated endpoint to an interior point of the Pesaran–Shin–Smith bracket. Throughout, $\tilde{k}$ is an organizing theoretical object rather than an estimated tuning parameter: the procedure never estimates it directly, and the bootstrap conditions implicitly on the realized trend content of the residualized statistic.

The contribution is threefold. First, the paper extends Pesaran–Shin–Smith bounds logic to a setting in which lagged levels are residualized against a high-dimensional nuisance space containing both stationary and integrated variables, producing a trend-absorption bracket: stationary controls leave the upper-bound logic intact, while integrated controls can move the statistic inside the interval by absorbing stochastic trends. Second, the paper connects this problem to double/debiased machine learning by showing how balanced first-stage projections, time-series cross-fitting, and sparse penalized residualization deliver an oracle-equivalent statistic under high-dimensional confounding.

Third, the paper explains why classical tabulated bounds values are not operationally valid in finite samples once generated-regressor error from high-dimensional residualization is present, and proposes a restricted *system* wild bootstrap that regenerates the dependent variable and the focal regressor jointly under the null, applying a common wild weight to the conditional and marginal residuals so that the contemporaneous correlation between them, the endogeneity channel of the Pesaran–Shin–Smith setting, is preserved in the bootstrap world. The bootstrap recomputes the full residualized statistic, holds the nuisance space fixed so as to condition on its realized trend content, regenerates the focal regressor from a marginal model that conditions on the differenced controls so that any common-trend component with the nuisance space is inherited from the fixed control path, and therefore estimates the relevant point inside the trend-absorption bracket.

To our knowledge, this is the first paper to develop a bounds-testing theory for cointegration in which high-dimensional residualization can itself change the number of stochastic trends governing the null distribution. The precise claim is not that DML and cointegration have never been combined, but that existing DML and high-dimensional time-series methods do not provide a bounds-testing theory in which residualization alters the effective number of stochastic trends entering the null.¹ The formal theory establishes oracle equivalence for high-dimensional stationary nuisance adjustment combined with fixed-dimensional integrated nuisance adjustment; the more general case of a growing integrated nuisance block is a sparse-cointegration selection problem and is left for future work.

The paper sits within the applied bounds-testing tradition that takes the finite-sample behavior of the test seriously. The same concern for usable critical values that motivates response-surface approximations to the Pesaran–Shin–Smith bounds (Kripfganz and Schneider, 2020) motivates the procedure here: when the conditioning set is large and persistent, the borrowed bounds are no longer the right reference, and a data-driven critical value computed by bootstrap is needed in their place. The contribution is therefore a practical inference tool for applied macroeconometric work in data-rich settings, not only a theoretical observation.

Table 1 locates the contribution against the literatures it draws on. The novelty is not any

¹This distinction separates the present paper from high-dimensional predictive regression with persistent regressors (Koo et al., 2020; Mei and Shi, 2024) and from sparse cointegration recovery (Gonzalo and Pitarakis, 2026). Those literatures inform the treatment of the integrated nuisance block, but they do not study ARDL-bounds inference after residualization.

single column but their conjunction: existing methods handle cointegration, or high-dimensional nuisance, but none treats the way residualization against an integrated nuisance space can change the number of stochastic trends that govern a bounds-test null.

Table 1: What is new: DML-Bounds against neighboring methods

Method	Cointegration bounds null	High-dim. nuisance	Trend absorption by residualization
Engle–Granger / Johansen	Yes	No	No
ARDL bounds (PSS)	Yes	No	No
Double/debiased ML	No	Yes	No
High-dim. time-series LASSO	Partial	Yes	Not for bounds inference
Sparse cointegration	Yes	Yes	Selects relation, not the bounds null
DML-Bounds (this paper)	Yes	Yes	Yes

Notes: “Trend absorption by residualization” refers to whether the method’s inference accounts for stochastic trends being absorbed when lagged levels are projected on a possibly-integrated nuisance space, the mechanism formalized here as the effective integrated count $\tilde{k}$.

The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 develops the procedure from the scalar error-correction root upward to the high-dimensional case. Section 4 details the DML-Bounds framework and penalty selection. Section 5 presents the asymptotic theory and primitive conditions. Section 6 explains why the classical bounds tables fail in finite samples and develops the bootstrap. Section 7 reports Monte Carlo evidence. Section 8 presents the empirical application. Section 9 concludes.

2 Related Literature

This paper connects three literatures that have largely evolved separately, and draws on two more for its high-dimensional and bootstrap components.

Cointegration, error correction, and single-equation bounds testing. The starting point is the classical cointegration literature, in which a stable long-run relation among integrated variables admits an error-correction representation (Engle and Granger, 1987). System approaches such as Johansen (1988, 1991) characterize cointegration through the rank of the long-run matrix; the present approach is single-equation, but the effective integrated count plays an analogous role after orthogonalization. The nonstandard Brownian-functional limits used here are closely related to the residual-based cointegration testing literature (Phillips and Ouliaris, 1990). Within the single-

equation tradition, [Banerjee et al. \(1998\)](#) test for a long-run relation through the error-correction coefficient under weak exogeneity, and [Pesaran et al. \(2001\)](#) develop the ARDL bounds test, which avoids pretesting by bracketing the null distribution between the cases in which the regressors are stationary and integrated. The single-equation tradition also includes efficient estimators of the long-run coefficient, fully modified OLS ([Phillips and Hansen, 1990](#)) and dynamic OLS ([Saikkonen, 1991](#); [Stock and Watson, 1993](#)); our object, however, is testing for the existence of a long-run relation after high-dimensional residualization, not efficient estimation of a known cointegrating vector. Recent response-surface work improves the finite-sample implementation of these bounds through critical-value regressions and approximate p -values ([Kripfganz and Schneider, 2020](#)). Our contribution begins from this single-equation bounds logic but asks what happens when the levels entering the test are first residualized against a high-dimensional nuisance space containing both stationary and integrated controls. The setting differs from [Kripfganz and Schneider \(2020\)](#) because the null distribution is altered not only by sample size and deterministic terms, but also by generated-regressor error from high-dimensional residualization.

Double machine learning and high-dimensional inference. The orthogonalization step follows the logic of double/debiased machine learning, which protects low-dimensional target parameters from first-stage regularization bias through Neyman orthogonality and cross-fitting ([Belloni et al., 2014](#); [Chernozhukov et al., 2018](#)). This literature is developed primarily for cross-sectional or stationary, weakly dependent data, where residualization does not alter the stochastic-trend structure of the target regressors. The present paper imports the orthogonalization and cross-fitting machinery but applies it in an integrated environment, where partialling out can change the very object, the number of surviving stochastic trends, that governs the test.

High-dimensional time-series LASSO. For stationary high-dimensional time series, oracle inequalities and valid post-selection inference have been developed by [Kock and Callot \(2015\)](#), [Medeiros and Mendes \(2016\)](#), and [Adamek et al. \(2023\)](#), with related penalized and factor-based methods for macroeconomic prediction in [Stock and Watson \(2002\)](#), [Smeekes and Wijler \(2018\)](#), and [Lee et al. \(2022\)](#). The first-stage prediction rates used in Section 5 are imported from this literature. Our setting differs because the tested regressors and part of the nuisance space may

be integrated, so these stationary rates apply only to the balanced, differenced first stage, while the integrated block is handled through super-consistent scaling rather than a standard restricted-eigenvalue condition.

LASSO with integrated and cointegrated regressors. The closest nonstationary work studies LASSO predictive regressions with persistent and potentially cointegrated predictors (Koo et al., 2020), and shows that the usual LASSO rates and restricted-eigenvalue arguments must be modified in the presence of unit-root regressors (Mei and Shi, 2024); the adaptive LASSO has also been studied for consistent model selection in nonstationary autoregressions (Kock, 2016). Most directly related, Gonzalo and Pitarakis (2026) use adaptive LASSO to identify sparse cointegrating relations and analyze its behavior under spurious integrated regressions. Our use of penalization is complementary rather than competing: we do not select the cointegrating relation itself, but use penalization, with adaptive weights, to prevent the nuisance projection from spuriously absorbing the stochastic trends that determine the bounds statistic. The problem here is not prediction with integrated predictors, nor recovery of a cointegrating vector, but valid bounds inference after residualizing a cointegration test against a high-dimensional mixed nuisance space.

Closely related recent work studies high-dimensional predictive regression with persistent and cointegrated regressors (Koo et al., 2020; Mei and Shi, 2024), sparse detection of cointegrating relations (Gonzalo and Pitarakis, 2026), and effective cointegration rank in high-dimensional unit-root systems (Fang et al., 2025). These papers are important because they show that regularization with integrated regressors is not a standard high-dimensional LASSO problem: irrelevant random walks can enter through spurious regression, and the usual restricted-eigenvalue and deviation arguments require modification. The present paper therefore takes a different route. In the formal theory, high-dimensionality is allowed in the stationary nuisance block, while the integrated nuisance block is kept fixed-dimensional. This avoids the separate sparse-cointegration selection problem and focuses the contribution on bounds inference after residualization. It also distinguishes the effective integrated count $\tilde{k}$ used here, the number of stochastic trends surviving residualization, from the effective cointegration rank of Fang et al. (2025), which is a reduced-rank object estimated directly from a high-dimensional unit-root system.

Bootstrap inference for nonstationary processes. Bootstrap inference with integrated processes is delicate because the null distribution depends on stochastic-trend functionals rather than standard Gaussian limits. We therefore follow the restricted-bootstrap logic developed for unit-root and cointegration-type settings (Cavaliere and Taylor, 2009; Smeekes, 2013). A second delicacy is specific to the Pesaran–Shin–Smith environment: the focal regressor need not be exogenous, so the null law of the bounds statistic depends on the contemporaneous correlation between the innovations of the regressor and the equation error. A wild bootstrap that holds the regressor fixed at its realized path and resamples only the conditional residual imposes independence between the two in the bootstrap world and therefore cannot replicate this correlation. The remedy, following the system bootstrap developed for cointegrated vector autoregressions by Cavaliere et al. (2012), is to regenerate the dependent variable and the focal regressor jointly, applying a common wild weight to the stacked conditional and marginal residuals so that their empirical cross-covariance, and any conditional heteroskedasticity, is carried into every bootstrap draw. We adopt this system logic in single-equation form and recompute the full residualized statistic so that generated-regressor error is included in the simulated null. The bootstrap is an implementation device here rather than the main theoretical object, so we rely on this established logic rather than developing a separate bootstrap invariance theory.

3 From a Single Equation to High-Dimensional Orthogonalization

The procedure is easiest to understand, and its asymptotics easiest to verify, by building it one ingredient at a time. This section starts from the scalar error-correction test, recovers the classical bounds test as the next case, and then introduces confounders one integration order at a time. The central object of the paper, a bracket over how much long-run trend the nuisance space absorbs, is already visible with a single integrated confounder. The high-dimensional procedure is then the natural generalization, and each high-level assumption of Section 5 is the limit of a case made explicit here.

3.1 Fundamentals: the scalar error-correction root

Let u_t be a candidate long-run equilibrium error and suppose it follows the AR(1) law

$$u_t = \rho u_{t-1} + e_t, \quad e_t \text{ i.i.d. } (0, \sigma^2). \quad (1)$$

The hypothesis of no cointegration is $H_0 : \rho = 1$, under which u_t is $I(1)$; the alternative $\rho < 1$ makes u_t stationary and represents a stable equilibrium. Writing $\pi = \rho - 1$ and $\Delta u_t = \pi u_{t-1} + e_t$, the test of $H_0 : \pi = 0$ is the Dickey–Fuller root of everything that follows. Its estimator $\hat{\pi} = \sum_t u_{t-1} \Delta u_t / \sum_t u_{t-1}^2$ obeys, under H_0 and with $\frac{1}{\sqrt{T}} u_{\lfloor rT \rfloor} \Rightarrow \sigma B(r)$,

$$T\hat{\pi} \Rightarrow \frac{\int_0^1 B dB}{\int_0^1 B^2}, \quad t_{\hat{\pi}} \Rightarrow \frac{\int_0^1 B dB}{\sqrt{\int_0^1 B^2}}, \quad (2)$$

the non-standard Dickey–Fuller limits, here in the no-deterministic form; under the unrestricted intercept of Case III the same functionals appear with B replaced by its demeaned version $B - \int_0^1 B$, which shifts the distribution without altering any of the arguments below. Two features carry through the entire construction. First, the rate is T , not $\sqrt{T}$: integrated regressors are super-consistent. Second, the limit is a functional of a Brownian motion, not a normal, which is exactly why critical values must be tabulated or simulated rather than read from a t table.

3.2 The bivariate bounds test

Now let the equilibrium error be unobserved and defined through two variables, $u_t = Y_t - \beta D_t$.

The conditional error-correction form is

$$\Delta Y_t = a_0 + \rho Y_{t-1} + \theta D_{t-1} + \delta \Delta D_t + \sum_i \gamma_i \Delta Y_{t-i} + e_t, \quad (3)$$

and the no-cointegration hypothesis is the joint restriction $H_0 : \rho = \theta = 0$ on the lagged levels $Z_{t-1} = (Y_{t-1}, D_{t-1})'$. The bounds-test F for H_0 has a limit that mixes a Dickey–Fuller component from Y_{t-1} with a Brownian component from D_{t-1} . Because the integration order of D is not known in advance, [Pesaran et al. \(2001\)](#) bracket it: the lower critical value treats D as $I(0)$, the upper value treats D as $I(1)$, and the realized F is read against the interval. The bracket is the response

to a single unknown, the integration order of the regressor. Everything below transplants that exact device to a new unknown.

Viewed through Frisch–Waugh–Lovell, (3) is a partialling-out problem: collect the short-run terms and deterministics in a nuisance vector and test the lagged levels after projecting them out. This is the seam along which the high-dimensional procedure cuts.

3.3 One stationary confounder: orthogonalization that does nothing

Add a single covariate w_t that is covariance-stationary, $I(0)$, and partial it out before testing. The orthogonalized regressor is $\tilde{Z}_{t-1} = Z_{t-1} - \hat{c} w_t$ with $\hat{c} = (\sum_t Z_{t-1} w_t) / (\sum_t w_t^2)$.

Lemma 1 (Stationary partialling preserves integration order). *Let w_t be $I(0)$ with $\sum_t w_t^2 = O_p(T)$ and Z_{t-1} be $I(1)$. Then the projection coefficient is $\hat{c} = O_p(1)$, a non-vanishing spurious-type quantity, because numerator and denominator are both $O_p(T)$. Nonetheless $w_t = O_p(1)$ at each fixed date, and under standard maximal-moment conditions $\max_{t \leq T} |w_t| = o_p(\sqrt{T})$, so $\hat{c} w_t = o_p(\sqrt{T})$ uniformly over the sample, and $\tilde{Z}_{t-1} = Z_{t-1} + o_p(\sqrt{T})$. Hence $\frac{1}{\sqrt{T}} \tilde{Z}_{[rT]} \Rightarrow \mathbf{B}(r)$, the same limit as the un-orthogonalized regressor.*

The content is elementary but decisive: a stationary regressor cannot track the stochastic trend of an integrated one, so the residual keeps its unit root. No matter how many stationary confounders are partialled out, the integrated variation that drives the bounds limit is untouched. In the language introduced next, a stationary nuisance leaves the effective integrated count at its maximum.

3.4 One integrated confounder: the bracket is born

Replace w_t by an integrated confounder, also $I(1)$. Now partialling can do real work, because two integrated series can share a stochastic trend. Two regimes arise.

If Z cointegrates with w , say $Y_{t-1} = \gamma w_{t-1} + (\text{stationary})$, then the population projection of Y_{t-1} on w_{t-1} recovers γ and the residual $\tilde{Y}_{t-1}$ is stationary. The trend has been absorbed. If instead Z and w are independent random walks, no population cointegrating coefficient exists; the sample projection coefficient converges to a non-degenerate random limit (spurious regression, Phillips, 1986), and the residual remains $I(1)$ but is contaminated in finite samples by the spurious

fit. The first regime stationarizes the regressor; the second leaves it integrated but invites spurious absorption that only regularization removes (Section 3.5).

This dichotomy is the whole idea, and it generalizes through the cointegrating rank.

Definition 1 (Effective integrated count). *Let $Z_{t-1} \in \mathbb{R}^k$ be $I(1)$ and let r be the number of independent cointegrating relations between Z and the integrated nuisance block reproduced by the population projection. The effective integrated count after orthogonalization is $\tilde{k} = k - r \in \{0, 1, \dots, k\}$.*

Remark 1 ($\tilde{k}$ is conceptual, not estimated). *The procedure does not require estimating $\tilde{k}$. Its role is theoretical: it indexes the asymptotic null experiment. Operationally, inference is obtained by recomputing the residualized statistic under the restricted system bootstrap of Algorithm 1, which holds the nuisance space at its realized path, regenerates the focal regressor from a marginal model that conditions on the differenced controls, so that its integration order and any common-trend component with the controls are preserved, and therefore conditions implicitly on the realized $\tilde{k}$ without ever estimating it as a tuning parameter.*

Proposition 1 (Trend-absorption bracket). *Suppose the orthogonalized regressors satisfy a functional central limit theorem on their $\tilde{k}$ surviving integrated directions. Then the orthogonalized bounds statistic converges to a bounds-type Brownian functional whose stochastic-trend dimension is $\tilde{k}$. In this sense residualization shifts the relevant null reference within the Pesaran–Shin–Smith bracket: $\tilde{k} = k$ places the limiting null at the upper ($I(1)$) endpoint, $\tilde{k} = 0$ at the lower ($I(0)$) endpoint, and $0 < \tilde{k} < k$ at an interior point. The statement is about the asymptotic null reference, not a claim that the finite-sample statistic is literally a classical Pesaran–Shin–Smith statistic; deterministic terms, lag structure, and generated-regressor error still affect the realized finite-sample law, which is why inference is obtained by bootstrap.*

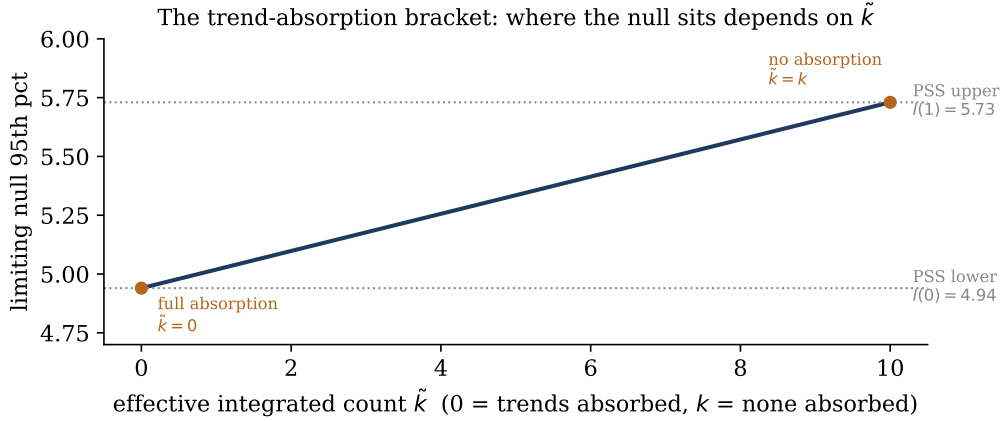


Figure 1: The trend-absorption bracket. As the effective integrated count $\tilde{k}$ falls from k (no absorption) to 0 (full absorption), the limiting null moves from the Pesaran–Shin–Smith upper bound to the lower bound. Classical bounds testing is the right endpoint; high-dimensional residualization can move the relevant null anywhere inside the interval.

Technical derivations for Lemma 1 and the trend-absorption rank argument are provided in Appendix A.

Remark 2 (Limit experiment versus finite-sample inference). *Proposition 1 is a statement about the limit experiment: the integrated-count logic organizes which Brownian functional governs the asymptotic null. It is not a claim that the realized statistic is a classical Pesaran–Shin–Smith statistic in finite samples. Once the lagged levels are residualized against high-dimensional nuisance estimates, generated-regressor error perturbs the finite-sample null, as the bootstrap evidence of Section 6 makes explicit. The bracket fixes the asymptotic reference; valid finite-sample inference is obtained by bootstrap.*

The two brackets of this section are one. In Section 3.2 the classical interval brackets the integration order of the regressor; here it brackets whether the nuisance space absorbed the trend. Both are instances of bracketing a single quantity, the number of tested regressors that carry a stochastic trend, which ranges over $\{0, \dots, k\}$ in either case. The Pesaran–Shin–Smith interval therefore remains the conceptual benchmark, because it brackets the number of stochastic trends carried by the tested regressors. In the high-dimensional setting, however, the realized finite-sample null is affected by generated-regressor error and must be obtained by bootstrap rather than read

directly from the classical tables (Section 6).

Two consequences are worth stating immediately. A purely stationary nuisance forces $r = 0$ and $\tilde{k} = k$ by Lemma 1, so the upper bound holds by construction and the bracket is inactive; this is why a classical low-dimensional ARDL, whose nuisance is lagged differences, sits at the upper endpoint. An integrated nuisance is what makes r unknown and the full interval live. And the construction carries a built-in identification caveat: if a confounder is cointegrated with the very relation under test, partialling it out drives $\tilde{k} \rightarrow 0$ and the relation is destroyed by the controls. The maintained requirement, made precise in Assumption 5, is that the nuisance space does not span the cointegrating relation of interest.

3.5 Generalization: high-dimensional mixed nuisance

Finally let the single confounder become a vector $W_t = (W'_{0t}, W'_{1t})'$ with $W_{0t} \in \mathbb{R}^{d_0}$ stationary, $W_{1t} \in \mathbb{R}^{d_1}$ integrated, $d = d_0 + d_1$, and d possibly comparable to or larger than T . Three features of the scalar analysis dictate the high-dimensional design.

Balance. The first-stage target ΔY_t is $I(0)$. By the logic of Lemma 1 in reverse, regressing a stationary target on integrated levels is unbalanced and spurious, so the integrated confounders must enter the ΔY projection in differences. The integrated levels enter only the projection of the lagged levels Z , which is where absorption lives.

Regularization. When the nuisance space is large, the unpenalized projection may be unstable, and in the presence of integrated controls it may also induce spurious trend absorption: an unpenalized projection of Z onto many independent integrated series spuriously stationarizes Z (Section 3.4). In implementation, ℓ_1 -penalization, with adaptive weights for the integrated block, is used as an empirical safeguard against over-absorbing stochastic trends. The formal theory below does not require selection consistency over a growing integrated block: high-dimensionality is allowed in the stationary nuisance block, while the integrated nuisance block is treated as fixed-dimensional. Thus adaptive penalization over integrated controls should be read as a diagnostic and stabilization device, not as a general sparse-cointegration selection theorem.

Cross-fitting. The nuisance is estimated on chronological training blocks separated from the evaluation block by an h -observation buffer, so that the first-stage error is decoupled from the evaluation-fold innovations. This turns the orthogonality requirement into a consequence of the

construction rather than an assumption (Lemma 2).

The effective integrated count of Definition 1 now ranges over $\{0, \dots, k\}$ according to the cointegrating rank between Z and the active integrated confounders, and Proposition 1 delivers the bracket. The remainder of the theory makes the limit precise and the inference operational.

3.6 Identification after residualization

The bracket carries an identification message that is economic, not only technical, and it is the central caveat of the method. Residualization is valid only when the controls remove nuisance stochastic trends, not the long-run equilibrium relation being tested. If a control variable is itself part of the equilibrium system, partialling it out absorbs the very trend that the test is meant to detect, driving $\tilde{k}$ toward zero and destroying the relation rather than cleaning it.

The consequence is a change in what is being tested. DML-Bounds does not test whether Y and D cointegrate unconditionally; it tests whether they cointegrate *conditional on the chosen nuisance space*. When the nuisance space is composed of variables genuinely distinct from the tested relation, the conditional and unconditional questions coincide and the procedure removes confounding. When the nuisance space overlaps the equilibrium system, the estimand shifts, and a non-rejection may reflect over-absorption rather than the absence of a long-run relation. This is the content of Assumption 5, stated here as interpretation: the maintained requirement is that the controls span confounding trends but not the cointegrating relation of interest. Because that requirement cannot be verified directly, the empirical work reports the full-versus-reduced control contrast as a diagnostic for when it may be failing, rather than relying on a single conditioning set.

4 Methodology: The DML-Bounds Framework

This section details the implementation of the procedure motivated in Section 3. Throughout, $Z_{t-1} = (Y_{t-1}, D_{t-1})'$ collects the lagged levels whose joint significance is tested, and W_t is the high-dimensional nuisance space. The procedure is DML-inspired rather than a direct application of standard double machine learning. In standard DML, residualization removes nuisance variation while preserving the low-dimensional target experiment. Here, because the tested regressors may be integrated, residualization can alter the stochastic-trend dimension of the target itself. This is why

the central theoretical object is not only an orthogonalized score but also the effective integrated count $\tilde{k}$: the orthogonality machinery is borrowed, but the limit theory is nonstandard because partialling out integrated controls changes the number of stochastic trends that govern the null, not merely the efficiency of the estimator.

4.1 Stage 1: balanced block cross-fitting and nuisance estimation

To isolate the long-run parameters from W_t , we use sequential block cross-fitting to prevent look-ahead bias and preserve temporal dependence. The sample is partitioned into K chronological blocks, with an h -observation buffer between training and evaluation samples to reduce dependence between them.

The first stage is balanced in integration order, as required by Section 3.5. The differenced target ΔY_t is $I(0)$ and is projected only on stationary regressors: stationary confounders in levels and integrated confounders in first differences. The lagged levels Z_{t-1} are $I(1)$ and are projected on the confounder levels. Using ℓ_1 -penalized LASSO, with adaptive weights for the integrated block in the sense of Zou (2006) and Fan and Li (2001), we estimate the out-of-fold conditional expectations

$$\hat{\ell}_{\Delta Y}(W_t) = \hat{\mathbb{E}}[\Delta Y_t \mid X_t], \quad (4)$$

$$\hat{m}_Y(W_t) = \hat{\mathbb{E}}[Y_{t-1} \mid W_{1t}, W_{0t}], \quad (5)$$

$$\hat{m}_D(W_t) = \hat{\mathbb{E}}[D_{t-1} \mid W_{1t}, W_{0t}], \quad (6)$$

where X_t collects the stationary balanced regressors. Following standard Post-LASSO practice, we refit by unpenalized OLS on the selected variables to reduce shrinkage bias. The adaptive weighting of the integrated block is important: vanilla ℓ_1 over-selects integrated regressors and thereby induces spurious trend absorption, whereas adaptive penalization curbs it. The validity result in Section 5 does not rely on consistent selection over a growing set of integrated controls. The integrated block is fixed-dimensional in the formal theorem; adaptive weighting is used in the empirical implementation to reduce spurious absorption and to support the trend-absorption diagnostic.

4.2 Stage 2: orthogonalization

Using the first-stage models, we construct the orthogonalized variables

$$\widetilde{\Delta Y}_t = \Delta Y_t - \widehat{\ell}_{\Delta Y}(W_t), \quad (7)$$

$$\widetilde{Y}_{t-1} = Y_{t-1} - \widehat{m}_Y(W_t), \quad (8)$$

$$\widetilde{D}_{t-1} = D_{t-1} - \widehat{m}_D(W_t), \quad (9)$$

and define $\widetilde{Z}_{t-1} = (\widetilde{Y}_{t-1}, \widetilde{D}_{t-1})'$.

4.3 Stage 3: the DML-Bounds statistic

The long-run parameters are estimated by unpenalized regression on the orthogonalized variables,

$$\widetilde{\Delta Y}_t = \widetilde{Z}_{t-1}' \hat{\beta} + \hat{e}_t, \quad \hat{\beta} = \left(\sum_t \widetilde{Z}_{t-1} \widetilde{Z}_{t-1}' \right)^{-1} \sum_t \widetilde{Z}_{t-1} \widetilde{\Delta Y}_t, \quad (10)$$

and a Wald- or F -type statistic is computed for $H_0 : \beta = 0$. Critical values are obtained by the bootstrap of Section 6 rather than from the classical tables.

4.4 Data-driven penalty selection

A critical implementation detail is the LASSO penalty λ . We use a data-driven rolling-origin Time-Series Cross-Validation (TSCV) mechanism. Let T_0 denote the minimum initial training sample. For origins $t = T_0, \dots, T-1$, the first-stage regressions are fitted on $\{1, \dots, t\}$, and λ_{opt} minimizes the average out-of-sample mean squared error,

$$\lambda_{opt} = \arg \min_{\lambda \in \Lambda} \frac{1}{T - T_0} \sum_{t=T_0}^{T-1} (\Delta Y_{t+1} - \widehat{\ell}_{\Delta Y}^{(\lambda)}(W_{t+1}))^2. \quad (11)$$

This balances bias and variance without violating temporal structure.

5 Asymptotic Theory

The plan mirrors the build. Section 5.2 states the high-level conditions under which the feasible statistic has the oracle bounds limit, with the trend-preservation condition cast as one endpoint of the bracket rather than a maintained hope. Section 5.3 verifies those conditions at the low level for the balanced Post-LASSO implementation: the orthogonality is derived, the compatibility condition is split by integration order, and the remainder is controlled. Section 5.4 assembles the equivalence theorem.

5.1 Fixed integrated nuisance block and high-dimensional stationary controls

The formal theory is developed for the empirically relevant case in which high-dimensionality arises from the stationary nuisance block, while the integrated nuisance block is fixed-dimensional. Write the nuisance vector as $W_t = (W'_{0t}, W'_{1t})'$, where $W_{0t} \in \mathbb{R}^{d_0}$ is covariance-stationary and potentially high-dimensional, with d_0 allowed to grow with T , while $W_{1t} \in \mathbb{R}^{d_1}$ is integrated with d_1 fixed. The lagged levels tested in the conditional ECM are collected in $Z_{t-1} \in \mathbb{R}^k$, with k fixed.

This restriction is deliberate, and it follows the lesson of the high-dimensional nonstationary regression literature (Koo et al., 2020; Mei and Shi, 2024; Gonzalo and Pitarakis, 2026). The stationary block is the natural domain for high-dimensional regularization: after differencing the integrated controls, the first-stage regression for ΔY_t is a stationary sparse prediction problem, and standard time-series LASSO rates apply. By contrast, selection among a growing number of integrated regressors is a separate sparse-cointegration problem, because irrelevant random walks may be selected through spurious regression and thereby falsely absorb stochastic trends, changing the effective integrated count that governs the bounds statistic. The present theory avoids that additional selection problem by keeping d_1 fixed and treating the integrated block as part of the low-dimensional long-run adjustment; the growing- d_1 case is left for future work.

Let the population projection of Z_{t-1} on the integrated nuisance block be $\Pi'W_{1t}$, and define the oracle residualized level vector $\dot{Z}_{t-1} = Z_{t-1} - \Pi'W_{1t}$. If Z_{t-1} and W_{1t} share r independent stochastic trends through this projection, the residualized level vector has effective integrated count $\tilde{k} = k - r$ in the sense of Definition 1. As emphasized there, $\tilde{k}$ is not estimated directly; it indexes the limit experiment, and inference is obtained operationally by recomputing the full residualized statistic

under the restricted bootstrap.

5.2 High-level conditions and the bracket

Let $\ell_{\Delta Y}(W_t) = \mathbb{E}[\Delta Y_t \mid W_t]$ and $m_Z(W_t) = \mathbb{E}[Z_{t-1} \mid W_t]$, and let $\tilde{Z}_{t-1} = Z_{t-1} - \hat{m}_Z(W_t)$, $\widetilde{\Delta Y}_t = \Delta Y_t - \hat{\ell}_{\Delta Y}(W_t)$.

Assumption 1 (First-stage prediction consistency). $\frac{1}{T} \sum_t (\hat{\ell}_{\Delta Y}(W_t) - \ell_{\Delta Y}(W_t))^2 = o_p(1)$ and $\frac{1}{T} \sum_t \|\hat{m}_Z(W_t) - m_Z(W_t)\|^2 = o_p(1)$.

Assumption 2 (Empirical remainder control). With $\delta_t = (\ell_{\Delta Y}(W_t) - \hat{\ell}_{\Delta Y}(W_t)) + (\hat{m}_Z(W_t) - m_Z(W_t))'\beta$, the integrated-scale interaction is negligible: $\frac{1}{T} \sum_t \tilde{Z}_{t-1} \delta_t = o_p(1)$.

Assumption 3 (Bracket endpoint: functional limit on the surviving trend). Under $H_0 : \beta = 0$, the $\tilde{k}$ integrated directions of the orthogonalized regressor satisfy $\frac{1}{\sqrt{T}} \tilde{Z}_{[rT]} \Rightarrow \mathbf{B}(r)$ with $\int_0^1 \mathbf{B} \mathbf{B}'$ almost surely positive definite on those directions. The case $\tilde{k} = k$ is the upper endpoint of Proposition 1.

Remark 3. Assumption 3 is no longer an assertion that orthogonalization spares the trend. It is the statement of one endpoint. The realized $\tilde{k}$, hence the position within the bracket, is determined by the data and is exactly what the bootstrap of Section 6 estimates.

5.3 Primitive verification under sparse linear nuisance

We now verify Assumptions 1–2 for the balanced Post-LASSO implementation, in the mixed design of Section 3.5.

Assumption 4 (Balanced first-stage specification). The target ΔY_t is $I(0)$ and is projected on the stationary set $X_t = (W'_{0t}, \Delta W'_{1t}, \Delta Y_{t-1}, \dots)'$, every component of which is covariance-stationary; the population first stage is the sparse linear projection $\ell_{\Delta Y}(W_t) = X'_t \gamma$ with active set of size s . Integrated levels W_{1t} enter only the projection m_Z of the lagged levels.

Assumption 5 (No nuisance redundancy with the tested relation). The nuisance space may share stochastic trends with Z , but it must not span the cointegrating relation whose stability is being tested. Equivalently, any trend absorption induced by W_{1t} is treated as nuisance adjustment only when it does not eliminate the equilibrium relation of interest.

This assumption marks the boundary of what the procedure can claim. DML-Bounds is not a high-dimensional cointegration detector that recovers a long-run relation regardless of the controls. It tests for a long-run relation after a specified nuisance adjustment. If the conditioning set contains variables that are themselves part of the equilibrium system, residualization may remove the relation rather than confounding, and the estimand changes. The trend-absorption diagnostic of Section 6 is a practical signal for this situation, not a formal guarantee against it, which is why the empirical application reports the full-versus-reduced contrast rather than a single verdict.

Assumption 6 (Innovations and weak convergence). *W_{0t} is covariance-stationary with uniformly bounded $(2 + \nu)$ moments and weak dependence sufficient for time-series LASSO prediction rates; $W_{1t} = W_{1,t-1} + \zeta_t$ with $\mathbb{E}[\zeta_t | \mathcal{F}_{t-1}] = 0$, finite second moments, and $\frac{1}{\sqrt{T}} \sum_{t \leq \lfloor rT \rfloor} \zeta_t \Rightarrow B_\zeta(r)$.*

Assumption 7 (Split compatibility). (i) *The centered stationary design X_t satisfies a restricted-eigenvalue condition on the sparse cone with probability approaching one, with $\frac{1}{T} \sum_t X_t X_t' \rightarrow_p \Sigma_X \succ 0$.* (ii) *For the integrated block it suffices that $\frac{1}{T^2} \sum_t W_{1t} W_{1t}' \Rightarrow \int_0^1 B_\zeta B_\zeta'$ is almost surely positive definite. No restricted-eigenvalue condition on the sparse cone is required for the integrated block.*

Assumption 8 (Growth and penalty). *$s \log d / \sqrt{T} \rightarrow 0$, and the penalty is large enough to regularize the noise while remaining small relative to the integrated scale; in implementation it is chosen by rolling-origin time-series cross-validation on a grid respecting this restriction.*

Remark 4 (Why the integrated restricted-eigenvalue problem dissolves). *Integrated regressors do not admit a deterministic restricted-eigenvalue bound: their scaled Gram matrix converges to the random $\int_0^1 B_\zeta B_\zeta'$, whose minimum eigenvalue is random rather than bounded below by a constant. Assumption 7 shows this is immaterial. Under H_0 the lagged-level error multiplies $\beta = 0$ and drops out (Section 5.4), so size is governed entirely by part (i), the stationary balanced design, where compatibility is standard. The integrated block needs only part (ii), the same non-degeneracy already used in the continuous-mapping step. The super-consistent T rate leaves enough slack that crude control suffices where the stationary block needs genuine compatibility.*

The orthogonality that earlier work imposed is now derived. Partition the sample into K contiguous blocks $\{I_k\}$ with h -buffers, and for $t \in I_k$ let $\hat{\gamma}^{(k)}$ be the Post-LASSO estimate from the complementary training fold, with error $r_t = X_t'(\hat{\gamma}^{(k)} - \gamma)$.

Lemma 2 (Cross-fit decoupling and stationary cross-moment control). *Under Assumptions 4–8, with $h \rightarrow \infty$, $h = o(T)$, the stationary Post-LASSO rate $\|\hat{\gamma}^{(k)} - \gamma\|_1 = O_p(s\sqrt{\log d/T})$, and the maximal cross-moment bound $\max_{1 \leq j \leq d} |\frac{1}{T} \sum_t (T^{-1/2} \tilde{Z}_{t-1}) X_{jt}| = O_p(\sqrt{\log d})$, then under H_0 ,*

$$\frac{1}{T} \sum_t \tilde{Z}_{t-1} r_t = O_p\left(\frac{s \log d}{\sqrt{T}}\right) = o_p(1).$$

Proof. Condition on the training fold $\mathcal{T}_k$; by the buffer and Assumption 6, $\hat{\gamma}^{(k)}$ is asymptotically independent of the evaluation-fold innovations, so its error c_k may be treated as fixed. Then $\|\frac{1}{T} \sum_{t \in I_k} \tilde{Z}_{t-1} r_t\| \leq \|G_k\|_\infty \|c_k\|_1$ with $G_k = \frac{1}{T} \sum_{t \in I_k} \tilde{Z}_{t-1} X_t'$. By Assumption 4 each column of X_t is stationary, so each entry of G_k is an integrated-times-stationary cross moment, $\frac{1}{T} \sum \tilde{Z}_{t-1} X_{jt} = \frac{1}{\sqrt{T}} \sum (\frac{1}{\sqrt{T}} \tilde{Z}_{t-1}) X_{jt} = O_p(1)$, and by the maintained maximal cross-moment bound $\|G_k\|_\infty = O_p(\sqrt{\log d})$ over the d coordinates. With $\|c_k\|_1 = O_p(s\sqrt{\log d/T})$ the product is $O_p(s \log d / \sqrt{T})$; summing the fixed K folds preserves the order. $\square$

Remark 5 (Contrast with an assumed orthogonality). *A conditional-mean-zero condition $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ fails once W is integrated, since $\mathbb{E}[W_{1t} \mid \mathcal{F}_{t-1}] = W_{1,t-1} \neq 0$. Lemma 2 replaces it with cross-fit decoupling, the balanced stationary basis, and a sup-norm cross-moment bound, at the cost of the marginally stronger growth condition $s \log d / \sqrt{T} \rightarrow 0$. The sup-norm bound $\|G_k\|_\infty = O_p(\sqrt{\log d})$ is imposed rather than derived from primitives: it is the high-dimensional analogue of the standard $O_p(1)$ bound on a single integrated-times-stationary cross moment, and holds under tail and mixing conditions on $(\tilde{Z}, X)$ through a maximal inequality of the type underlying the high-dimensional time-series bounds of [Adamek et al. \(2023\)](#). If the maximal cross moment were controlled only at a higher polylogarithmic rate $(\log d)^a$ with $a > \frac{1}{2}$, the argument goes through unchanged with the growth condition strengthened to $s(\log d)^{a+\frac{1}{2}} / \sqrt{T} \rightarrow 0$.*

Proposition 2 (Primitive remainder control). *Under Assumptions 4–8 and the buffer condition, $\frac{1}{T} \sum_t \tilde{Z}_{t-1} \delta_t = O_p(s \log d / \sqrt{T}) = o_p(1)$ under H_0 , verifying Assumption 2.*

Proof. Under H_0 the lagged-level error multiplies $\beta = 0$, so $\delta_t = r_t$; apply Lemma 2. $\square$

5.4 The theorem ladder

The asymptotic argument is organized as a sequence. Proposition 3 establishes the limit of the infeasible oracle statistic, indexed by the effective integrated count. Proposition 4 shows, at a high level, that the feasible cross-fitted statistic is asymptotically equivalent to the oracle, and Theorem 1 establishes this equivalence at the primitive level under the fixed-integrated-block design, the paper's main formal result. Algorithm 1 gives the restricted system wild-bootstrap implementation that inference uses in finite samples; Theorem 2 establishes its consistency in the oracle fixed-integrated-block case, with scope and supporting evidence stated in Remark 6. Corollary 2 records the classical bounds test as the special cases of the bracket.

For the partially linear conditional ECM $\Delta Y_t = Z'_{t-1}\beta + g(W_t) + \epsilon_t$ with $\mathbb{E}[\epsilon_t | Z_{t-1}, W_t] = 0$, the population projection gives $\ell_{\Delta Y}(W_t) = m_Z(W_t)'\beta + g(W_t)$.

Proposition 3 (Oracle residualized bounds limit). *Let $\dot{Z}_{t-1} = Z_{t-1} - m_Z(W_t)$ and $\dot{\Delta Y}_t = \Delta Y_t - \ell_{\Delta Y}(W_t)$ be the infeasible oracle residuals, and let $\hat{\beta}^*$ be the oracle estimator from regressing $\dot{\Delta Y}_t$ on $\dot{Z}_{t-1}$. Under Assumption 3, with $\tilde{k}$ the effective integrated count of Definition 1,*

$$T(\hat{\beta}^* - \beta) \Rightarrow \left(\int_0^1 \mathbf{B}\mathbf{B}' \right)^{-1} \left(\int_0^1 \mathbf{B} dM \right)$$

on the $\tilde{k}$ surviving integrated directions, and the oracle Wald-/F-type bounds statistic converges to a bounds-type Brownian functional indexed by $\tilde{k}$. This is the core limit: residualization changes the stochastic-trend dimension that governs the test from k to $\tilde{k}$.

Proof. With $\beta = 0$ under the null, $\dot{\Delta Y}_t = \epsilon_t$. By Assumption 3 and the continuous mapping theorem, $\frac{1}{T^2} \sum_t \dot{Z}_{t-1} \dot{Z}'_{t-1} \Rightarrow \int_0^1 \mathbf{B}\mathbf{B}'$, almost surely invertible on the $\tilde{k}$ surviving directions, and $\frac{1}{T} \sum_t \dot{Z}_{t-1} \epsilon_t \Rightarrow \int_0^1 \mathbf{B} dM$ for the martingale limit M of the partial-sum process of ϵ_t . The bounds statistic is a continuous functional of these quadratic forms, hence converges to the Pesaran–Shin–Smith functional with $\tilde{k}$ integrated directions. $\square$

The feasible statistic replaces the unknown projections by cross-fitted Post-LASSO estimates. Substituting and subtracting the feasible lagged projection yields $\widetilde{\Delta Y}_t - \tilde{Z}'_{t-1}\beta = \epsilon_t + \delta_t$, and under

H_0 the lagged-level component of δ_t drops out, so

$$T(\hat{\beta} - \beta) = \left(\frac{1}{T^2} \sum_t \tilde{Z}_{t-1} \tilde{Z}'_{t-1} \right)^{-1} \left[\frac{1}{T} \sum_t \tilde{Z}_{t-1} \epsilon_t + \frac{1}{T} \sum_t \tilde{Z}_{t-1} \delta_t \right]. \quad (12)$$

Proposition 4 (Feasible DML statistic equals oracle statistic). *Under Assumptions 1–3 and the primitive conditions of Section 5.3, the feasible DML-Bounds statistic differs from the oracle statistic of Proposition 3 by $o_p(1)$, and therefore shares its limit experiment, a Pesaran–Shin–Smith functional indexed by $\tilde{k}$.*

Proof. By Proposition 2 (equivalently Assumption 2) the second bracket term in (12) is $o_p(1)$, so the feasible parameter expansion coincides with the oracle expansion of Proposition 3 to first order. The cross-fit decoupling of Lemma 2 is what makes the generated-regressor interaction negligible at the integrated scale. The bounds statistic, a continuous functional of the coefficient process and the associated quadratic forms, inherits the oracle limit. $\square$

The next theorem states this equivalence at the primitive level under the fixed-integrated-block design of Section 5.1, replacing the high-level remainder condition by explicit rate and sparsity primitives. It is the sharpest form of the result and the one we regard as the formal contribution.

Theorem 1 (Feasible–oracle equivalence with fixed integrated nuisance). *Let $Z_{t-1} \in \mathbb{R}^k$ be the lagged level vector entering the bounds statistic, with k fixed, and let $W_t = (W'_{0t}, W'_{1t})'$ with $W_{0t} \in \mathbb{R}^{d_0}$ covariance-stationary and potentially high-dimensional and $W_{1t} \in \mathbb{R}^{d_1}$ integrated with d_1 fixed. Suppose Z_{t-1} and W_{1t} share r cointegrating relations, so that the oracle residual $\dot{Z}_{t-1} = Z_{t-1} - \Pi' W_{1t}$ has effective integrated count $\tilde{k} = k - r$. Assume: (i) the innovations driving W_{1t} and the disturbance ϵ_t satisfy a joint functional central limit theorem; (ii) ϵ_t is a martingale difference sequence with finite $2 + \nu$ moments; (iii) the stationary first stage is s_0 -sparse and estimated by h -block cross-fitted Post-LASSO; (iv) the stationary Post-LASSO prediction error satisfies the time-series rate $\frac{1}{T} \sum_t (\hat{\ell}_{\Delta Y}(W_t) - \ell_{\Delta Y}(W_t))^2 = O_p(s_0 \log d_0 / T)$; (v) $s_0 \log d_0 / \sqrt{T} \rightarrow 0$; and (vi) K is fixed while the buffer satisfies $h \rightarrow \infty$, $h = o(T)$. Then, under the null of no conditional cointegration,*

$$\frac{1}{T^2} \sum_t \tilde{Z}_{t-1} \tilde{Z}'_{t-1} - \frac{1}{T^2} \sum_t \dot{Z}_{t-1} \dot{Z}'_{t-1} = o_p(1), \quad \frac{1}{T} \sum_t \tilde{Z}_{t-1} \epsilon_t - \frac{1}{T} \sum_t \dot{Z}_{t-1} \epsilon_t = o_p(1),$$

and consequently the feasible DML-Bounds statistic converges to the same bounds-type Brownian

functional as the oracle statistic, indexed by $\tilde{k}$. The stochastic-trend dimension entering the bounds null is therefore not the original k , but the surviving dimension $\tilde{k} = k - r$.

Proof. Decompose the feasible residual as $\tilde{Z}_{t-1} = \dot{Z}_{t-1} - (\hat{\Pi} - \Pi)'W_{1t} - \hat{r}_{Zt}$, where $(\hat{\Pi} - \Pi)'W_{1t}$ is the estimation error of the integrated projection and $\hat{r}_{Zt}$ collects the stationary first-stage error. Because d_1 is fixed, $\hat{\Pi}$ is a fixed-dimensional cointegrating projection and standard cointegration theory gives $\|\hat{\Pi} - \Pi\| = O_p(T^{-1})$, while $\frac{1}{T} \sum_t W_{1t}\epsilon_t = O_p(1)$ by (i). Hence the score perturbation from the integrated projection is $(\hat{\Pi} - \Pi)'(\frac{1}{T} \sum_t W_{1t}\epsilon_t) = O_p(T^{-1})O_p(1) = o_p(1)$. The remaining perturbation is the stationary high-dimensional term. The first-stage target is stationary and the regressors are balanced in integration order, so the time-series Post-LASSO rate (iv) applies; with h -block cross-fitting the estimation error is decoupled from the evaluation-fold innovations (Lemma 2), which replaces the Cauchy–Schwarz bound by a mean-zero cross-fold sum, so that $\frac{1}{T} \sum_t \dot{Z}_{t-1}\hat{r}_{Zt} = O_p(s_0 \log d_0/\sqrt{T}) = o_p(1)$ under (v). Both objects entering the statistic are therefore asymptotically equivalent to their oracle counterparts. On the $\tilde{k}$ surviving integrated directions the oracle residual obeys the invariance principle of Assumption 3, $T^{-1/2}\dot{Z}_{\lfloor sT \rfloor} \Rightarrow B(s)$, so $T^{-2} \sum_t \dot{Z}_{t-1}\dot{Z}_{t-1}' \Rightarrow \int_0^1 BB'$ and $T^{-1} \sum_t \dot{Z}_{t-1}\epsilon_t \Rightarrow \int_0^1 B dM$. The bounds statistic is a continuous functional of these quadratic forms, and the continuous mapping theorem delivers the bounds-type Brownian limit indexed by $\tilde{k}$. $\square$

Propositions 3 and 4 identify the asymptotic null. They do not, on their own, deliver finite-sample critical values, because the $O_p(s \log d/\sqrt{T})$ remainder is non-negligible at empirically relevant sample sizes (Section 6). The Brownian-functional derivation and the feasible-oracle expansion are collected in Appendix A. Inference therefore rests on the bootstrap, and the result it requires is the following.

Algorithm 1 (Restricted system wild-bootstrap inference). Given the data and the imposed null H_0 , compute the feasible DML-Bounds statistic F as in Section 6, and estimate two auxiliary models: the restricted conditional model for ΔY_t under H_0 , with residuals $\hat{\epsilon}_t$, and a marginal model for ΔD_t on its own lags and the differenced controls, with residuals $\hat{v}_t$. For $b = 1, \dots, B$: (i) draw a single Rademacher sequence $\eta_t \in \{-1, +1\}$ and apply it to the *stacked* residual vector, $(\epsilon_t^*, v_t^*) = (\hat{\epsilon}_t \eta_t, \hat{v}_t \eta_t)$, so that the empirical contemporaneous covariance between the conditional and marginal innovations is preserved in every draw; (ii) regenerate $\Delta D^{*(b)}$ from the marginal

dynamics with the control path held fixed, cumulate to the level path $D^{*(b)}$, then regenerate $\Delta Y^{*(b)}$ under the restricted null dynamics driven by $\Delta D^{*(b)}$ and ϵ^* , and cumulate to $Y^{*(b)}$; the nuisance space W is held fixed at its realized path; (iii) recompute the *entire* residualized statistic $F^{*(b)}$ on $(Y^{*(b)}, D^{*(b)}, W)$, re-selecting the first-stage level supports on the regenerated paths so that selection error is reflected in the bootstrap law. The bootstrap critical value is the $1 - \alpha$ quantile of $\{F^{*(b)}\}_{b=1}^B$ and the bootstrap p -value is $B^{-1} \sum_b \mathbf{1}\{F^{*(b)} \geq F\}$. When the focal regressor is maintained to be strongly exogenous, holding D fixed at its realized path in step (ii) recovers the fixed-regressor scheme as a special case.

The next result is the theoretical anchor for Algorithm 1. It establishes consistency of the system bootstrap in the setting where the rest of the formal theory lives, the fixed-integrated-block design of Section 5 with oracle nuisance projections, so that the one mechanism the bootstrap must reproduce, the joint null law of the conditional and marginal innovations and the stochastic trends they drive, is isolated from first-stage estimation error.

Theorem 2 (Bootstrap validity in the oracle fixed-integrated-block case). *Let the conditions of Theorem 1 hold with the nuisance projections known (oracle m_Z and $\ell_{\Delta Y}$), and let v_t denote the innovation of the marginal process for ΔD_t , the population counterpart of $\hat{v}_t$ in Algorithm 1 (the innovations of the integrated nuisance block are denoted ζ_t throughout and play no direct role here, as W is held fixed). Assume in addition that (i) (ϵ_t, v_t) is a strictly stationary martingale difference array with respect to the joint filtration, with finite fourth moments and contemporaneous covariance matrix Σ , unrestricted off the diagonal, so that endogeneity of the focal regressor is permitted; (ii) under H_0 the restricted conditional and marginal coefficient estimates entering Algorithm 1 are consistent; and (iii) the null limit distribution of the DML-Bounds statistic F , the $\tilde{k}$ -indexed bounds functional of Theorem 1, has a continuous distribution function. Then, conditionally on the data,*

$$F^* \Rightarrow_p \text{ the same } \tilde{k}\text{-indexed bounds functional,} \quad \text{and} \quad \sup_x |\mathbb{P}^*(F^* \leq x) - \mathbb{P}(F \leq x)| \rightarrow_p 0$$

under H_0 , so the bootstrap critical values of Algorithm 1 deliver asymptotic size α and the bootstrap p -value is asymptotically uniform. The result holds for any value of the contemporaneous correlation in Σ ; the fixed-regressor scheme obtains the same conclusion only when that correlation is zero, because its simulated innovations are independent of the fixed regressor path by construction. The

key steps are recorded in Appendix A.5.

The theorem is deliberately scoped to oracle projections: it isolates bootstrap validity from first-stage estimation. The bridge to the feasible procedure is the bootstrap analogue of the remainder control already used on the sample side, and it completes the theorem ladder.

Assumption 9 (Bootstrap first-stage negligibility). *Let $\tilde{Z}_{t-1}^*$ and δ_t^* denote the bootstrap-path counterparts of the residualized regressors and the first-stage remainder of Assumption 2, computed on (Y^*, D^*, W) with the supports re-selected as in Algorithm 1. Then, conditionally on the data, $\frac{1}{T} \sum_t \tilde{Z}_{t-1}^* \delta_t^* = o_{p^*}(1)$ in probability.*

Corollary 1 (Feasible bootstrap validity). *Under the conditions of Theorem 2 and Assumption 9, the conclusion of Theorem 2 extends to the feasible statistic: the feasible bootstrap distribution is asymptotically equivalent, conditionally in probability, to the oracle bootstrap distribution, and hence $\sup_x |\mathbb{P}^*(F^* \leq x) - \mathbb{P}(F \leq x)| \rightarrow_p 0$ for the feasible DML-Bounds statistic under H_0 .*

The proof is immediate from the structure already in place: on each bootstrap path the feasible statistic differs from the oracle bootstrap statistic by the bootstrap remainder term, which Assumption 9 makes negligible, so the two share the conditional limit of Theorem 2; this is the bootstrap-side mirror of Proposition 4. Assumption 9 is a high-level condition, stated rather than derived from primitives: it is the bootstrap analogue of Assumption 2, and the same mechanisms that deliver the sample-side version, the balanced stationary first stage, cross-fit decoupling, and the Post-LASSO rate of Lemma 2, are the natural route to it, since the bootstrap paths are generated from restricted dynamics whose innovations satisfy the same moment structure. Verifying it from primitives uniformly over bootstrap paths, and over general first-stage learners, is the invariance problem left to future work; the per-path support re-selection of Algorithm 1 is designed precisely so that the bootstrap law reflects the selection error the condition controls, and the simulations of Section 7 evaluate the assumption’s adequacy directly. The theorem ladder is then complete at three rungs: primitive first-stage conditions imply feasible–oracle equivalence of the statistic (Theorem 1); the oracle bootstrap is valid under arbitrary endogeneity correlation (Theorem 2); and bootstrap-side first-stage negligibility transfers that validity to the feasible procedure (Corollary 1).

Remark 6 (Validity: motivation, scope, and evidence). *Algorithm 1 is motivated by Propositions 3 and 4, which identify the oracle null law and the feasible-oracle equivalence, and by existing bootstrap*

theory for integrated processes (Cavaliere and Taylor, 2009; Smeekees, 2013) and for cointegrated systems (Cavaliere et al., 2012). Three design choices carry the argument. First, the common wild weight on the stacked residuals preserves the contemporaneous correlation between the marginal innovation of D and the conditional error, which is what a fixed-regressor scheme destroys and what the endogeneity-robust Pesaran–Shin–Smith setting requires. Second, holding W fixed conditions on the realized trend content of the nuisance space, the object the trend-absorption bracket is built on. Third, because the marginal model for ΔD conditions on the differenced controls, any stochastic trend that D shares with the nuisance space enters D^* through the fixed control path, so the realized cointegration configuration between the focal regressor and the controls, and hence the realized $\tilde{k}$, is inherited by the bootstrap world rather than broken by independent regeneration. A fully general invariance theorem for arbitrary learners and arbitrary mixed integrated nuisance spaces is beyond the scope of this paper. Its finite-sample performance is instead evaluated directly: the Monte Carlo experiments of Section 7, run in the designs that match the mechanisms studied here, show that the procedure restores empirical size to nominal levels across the integrated-nuisance designs where the classical bound over-rejects, retains useful power, and controls size under strong contemporaneous correlation between the regressor innovations and the equation error without relying on the conditional specification to absorb it (Section 7.4). Theorem 2 closes part of the gap formally: in the oracle fixed-integrated-block design the system bootstrap is consistent for any value of the endogeneity correlation, so what remains for future work is uniformity over first-stage learners, not the bootstrap principle itself.

Corollary 2 (Classical bounds as special cases). *Within the limit experiment of Proposition 4: (i) if no integrated nuisance trend is partialled out, $r = 0$ and $\tilde{k} = k$, and the statistic has the Pesaran–Shin–Smith upper-bound limit; (ii) if all tested stochastic trends are absorbed, $r = k$ and $\tilde{k} = 0$, and the statistic has the lower-bound ($I(0)$) limit; (iii) for $0 < \tilde{k} < k$ the limit is an interior point of the Pesaran–Shin–Smith bracket. The classical bounds test of Pesaran et al. (2001) is therefore the $d_1 = 0$ corner of the framework, in which the nuisance space carries no integrated component and $\tilde{k} = k$ holds by Lemma 1.*

Remark 7 (Validity of h -block cross-fitting). *With K fixed and $T \rightarrow \infty$ each block has length $O(T)$ and the un-orthogonalized integrated regressors satisfy the FCLT within blocks. The buffer*

discards at most $h(K - 1)$ observations; if $h = o(T)$ then $h(K - 1)/T \rightarrow 0$ and the Riemann-sum approximations behind the Brownian functionals are unaffected. This is a first-order justification, not a separate invariance theorem.

Remark 8 (Scope of the verification). *Propositions 3 and 4 are established under the stated conditions. The bootstrap of Algorithm 1 is motivated by these results and by existing bootstrap theory for integrated processes, and is evaluated directly in the simulations of Section 7. Two inputs are imported from the standard literature rather than re-derived: the functional limit on the surviving integrated directions (Assumption 3), verified by hand in the scalar cases of Sections 3.3–3.4, and the stationary Post-LASSO rate used in Lemma 2 (Kock and Callot, 2015; Medeiros and Mendes, 2016). The paper’s contribution is the identification of $\tilde{k}$ as the object governing the test, the reduction of the high-dimensional integrated problem to these standard inputs, and the bracket that organizes them.*

6 Inference: Why the Bounds Tables Fail and How to Replace Them

Proposition 4 says the feasible statistic shares the limit of a classical bounds test in which $\tilde{k}$ of the k tested regressors are integrated. It does *not* say the realized statistic can be read against the tabulated values $I(0) = 4.94$, $I(1) = 5.73$ once the nuisance space is high-dimensional and integrated. The reason is the rate, not a defect of the theorem. The remainder of Proposition 2 is $O_p(s \log d / \sqrt{T})$, which vanishes only in the limit. At the sample sizes applied work actually uses it is order one: with $s \approx 3$, $d \approx 40$, $T = 200$ it equals $3 \log 40 / \sqrt{200} \approx 0.78$, and closer to 0.9 once the differenced block is counted in d . The first-order asymptotics have therefore not taken hold, the generated-regressor term is still inflating the null, and 5.73 is the not-yet-reached asymptotic reference rather than a valid finite-sample one. In the Monte Carlo designs of Section 7, comparison against 5.73 can severely over-reject under the null, with empirical rejection rates far above the nominal level and the empirical 95% quantile of the statistic several times the borrowed bound.

The remedy is to generate the critical value from the data, which is also the data-driven completion of the bracket: rather than read against the fixed interval, read against a null distribution constructed from the actual regressor configuration, so that the realized $\tilde{k}$ enters automatically.

Crucially, it is Proposition 4 that licenses this. The theorem establishes that the feasible statistic has a well-defined, non-degenerate limit that is a continuous functional of the same Brownian functionals as the classical test. That non-degeneracy and continuity are the conditions under which a restricted wild bootstrap can be expected to approximate the relevant finite-sample null: the bootstrap is designed to do so by holding the nuisance space at its realized path, regenerating the focal regressor in a way that preserves its unit root and its trend links to the controls, carrying the endogeneity correlation into every draw through the common wild weight, and recomputing the full residualized statistic, including first-stage selection error. Proposition 4 identifies the object the bootstrap must reproduce. The bootstrap is theoretically grounded at two levels: Theorem 2 establishes its consistency in the oracle fixed-integrated-block design for any value of the endogeneity correlation, and its finite-sample behavior with estimated first stages is assessed directly by simulation; in this sense the construction rests on standard wild-bootstrap arguments for integrated data and cointegrated systems (Cavaliere and Taylor, 2009; Cavaliere et al., 2012; Smeekes, 2013), with general-learner uniformity left to future work. The asymptotic theory is thus the foundation of the inference, not an ornament beside it. The bootstrap recomputes the full statistic, first-stage noise included, so it mimics the finite-sample null that the $O_p(s \log d / \sqrt{T})$ term shows the asymptotics have not yet reached.

The procedure is the one stated compactly as Algorithm 1; we restate it here in the operational form used in the empirical specification, with the restricted null and the regeneration step written out.

Restricted system wild bootstrap (Algorithm 1, applied form). (1) Compute the observed statistic F . (2) Estimate the restricted *conditional* model under H_0 of no cointegration using the same deterministic terms and short-run lag structure as the empirical specification, but excluding the lagged levels and excluding the high-dimensional confounder levels, obtaining residuals $\hat{\epsilon}_t$; estimate the *marginal* model for the focal regressor, ΔD_t on an intercept, its own lags, and the differenced controls, obtaining residuals $\hat{v}_t$. (3) For $b = 1, \dots, B$, draw a single Rademacher sequence $\eta_t \in \{-1, +1\}$ and form the jointly weighted innovations $\epsilon_t^* = \hat{\epsilon}_t \eta_t$ and $v_t^* = \hat{v}_t \eta_t$; regenerate ΔD_t^* recursively from the marginal dynamics with the control path fixed, cumulate

to D_t^* ; regenerate ΔY_t^* recursively from the restricted conditional dynamics driven by ΔD_t^* and ϵ_t^* , cumulate to Y_t^* ; and recompute the DML-Bounds statistic F^* on (Y^*, D^*, W) with W held fixed, re-selecting the level supports on the regenerated paths.

(4) Reject at level α if F exceeds the $1-\alpha$ quantile of $\{F^*\}$; report the bootstrap p -value $B^{-1} \sum_b \mathbf{1}\{F_b^* \geq F\}$.

Endogenous regressors and why the system form is not optional. The Pesaran–Shin–Smith framework is built for the case in which the focal regressor is not exogenous: the conditional error-correction model exists precisely because the innovations v_t of the marginal process for ΔD_t may be contemporaneously correlated with the disturbance of the structural equation, and the conditional specification absorbs that correlation through the contemporaneous ΔD_t term. An earlier version of this procedure resampled only $\hat{\epsilon}_t$ and held D fixed at its realized path. That scheme has a defect that is invisible under exogeneity but decisive without it: because the wild weights multiply $\hat{\epsilon}_t$ alone, the bootstrap innovations are independent of the fixed regressor path by construction, so the simulated null corresponds to a world in which $\text{corr}(\epsilon_t, v_t) = 0$ regardless of the correlation in the data. The statistic’s null law, however, involves cross-functionals of the Brownian motions driven by (ϵ_t, v_t) , and after high-dimensional residualization the generated-regressor terms interact with that dependence. A fixed-regressor wild bootstrap therefore cannot be relied upon to replicate the endogeneity structure of the data-generating process: any adequacy it displays rests entirely on the contemporaneous ΔD_t term of the restricted specification having absorbed the dependence before resampling, a knife-edge property of the conditional model rather than of the resampling scheme, as the simulations of Section 7.4 make precise. The system form of Algorithm 1 repairs this at essentially no cost: the same wild weight multiplies the stacked residual pair, so every bootstrap draw carries the empirical cross-covariance of $(\hat{\epsilon}_t, \hat{v}_t)$, and with Rademacher weights the conditional heteroskedasticity of both innovations is preserved as well. Under exogeneity the two schemes coincide asymptotically, so nothing is lost by adopting the system form as the default.

The system regeneration preserves what the bracket needs preserved. Cumulating v^* keeps D^* integrated by construction, so the tested regressor’s stochastic trend survives every draw; holding W fixed conditions on the realized trend content of the nuisance space; and because the marginal model conditions on the differenced controls, any common-trend component between D and the

integrated controls is transmitted to D^* through the fixed control path rather than destroyed by independent resampling. This is precisely what makes the bootstrap distribution track the correct point inside the bracket. The empirical application of Section 8 is therefore reported against the bootstrap distribution, not against the classical bounds, and the two can disagree: a statistic comfortably above 5.73 may sit below the bootstrap critical value once the integrated nuisance is accounted for.

Definition 2 (Trend-absorption diagnostic). *Let p^{ad} and p^{ols} be the bootstrap p -values of the DML-Bounds statistic under the adaptive and the unpenalized m_Z projection, and let p^{full} and p^{red} be the bootstrap p -values under the full nuisance set and under a reduced set that omits the controls most likely to be cointegrated with the tested relation. The trend-absorption diagnostic is the pair of gaps $\Delta_m = p^{\text{ols}} - p^{\text{ad}}$ and $\Delta_W = p^{\text{full}} - p^{\text{red}}$, read together with the stability of the long-run coefficient across the four fits.*

Remark 9 (Operational rule). *A large positive Δ_W , a verdict that rejects under the reduced set but not under the full set, together with a long-run coefficient that is more stable and more sharply estimated under the reduced or adaptive fit, is evidence that the nuisance space is absorbing part of the tested relation, that is, that Assumption 5 is locally violated; the reduced-set verdict is then the more credible one. Concordant verdicts across all four fits, by contrast, indicate that the conclusion is not an artifact of over-absorption. Because vanilla ℓ_1 over-selects integrated regressors and thereby induces spurious absorption, the adaptive projection is the default; the unpenalized projection is retained only as the Δ_m comparison. The diagnostic is itself a hypothesis-generating device, not a formal test, and the empirical application of Section 8 reports it in full.*

7 Simulation Evidence

This section evaluates the finite-sample performance of the DML-Bounds procedure in high-dimensional environments. The designs are organized along the two axes that the theory uses: the dimensional regime (d small, d comparable to T , and $d > T$) and the integration content of the nuisance space (stationary, mixed, and integrated). The aim is not to claim universal dominance over classical ARDL bounds testing, but to assess whether the orthogonalized implementation

provides a feasible and stable alternative when the nuisance space is too large or too collinear for unpenalized projection, and to confirm the inference claims of Section 6.

7.1 Design grid

The experiments are organized as a grid of five designs that isolate the mechanisms of Section 3, reported in Table 2. Design A validates that the procedure reproduces classical Pesaran–Shin–Smith behavior in the low-dimensional limit. Design B is the high-dimensional stationary case in which residualization should help relative to a naive specification. Designs C and D are the heart of the paper: integrated and then cointegrated nuisance controls, where the effective integrated count $\tilde{k}$ falls below k and classical bounds inference is distorted by trend absorption. Design E stresses the procedure under weak signal, near unit roots, and small samples. Every cell is reported on the classical bound, the empirical critical value, and the bootstrap, and where relevant on the accuracy with which the implied trend absorption is recovered, with and without cross-fitting, and under adaptive versus plain LASSO.

Table 2: Monte Carlo Design Grid

Design	Nuisance space	$\tilde{k}$ vs k	Purpose
A	none / low-dim stationary	$\tilde{k} = k$	reproduce PSS behavior
B	high-dim $I(0)$	$\tilde{k} = k$	residualization improves size/power
C	high-dim with $I(1)$ block	$\tilde{k} \leq k$	classical bounds distorted
D	cointegrated $I(1)$ controls	$\tilde{k} < k$	trend absorption reduces dimension
E	weak signal, near unit root, small T	varies	robustness

Notes: Sample sizes $T \in \{80, 120, 200\}$ and dimension ratios giving $d \in \{50, 100, 200\}$ are used in Design E. For each cell the replication code reports empirical size, power, rejection rates under the classical lower and upper bounds, rejection rates under the bootstrap, recovery of the implied trend absorption, the effect of cross-fitting versus none, and the effect of adaptive versus ordinary LASSO. The main results below report these designs under the feasible restricted wild bootstrap; a complementary fixed-dimension sensitivity study follows.

7.2 Main results: bootstrap size and power across designs

Table 3 reports the headline evidence: empirical size against the borrowed 5.73 bound, empirical size against the feasible restricted wild bootstrap, and bootstrap power, for each design. This is the experiment that speaks directly to the paper’s claim, because the inference reference is the bootstrap that the method actually uses, not an infeasible oracle benchmark. Monte Carlo design details,

bootstrap settings, the random seed, and implementation choices are reported in Appendix B.

Table 3: Design A–E Results

Design	T	d	rej @ 5.73 (size)	rej @ boot (size)	power @ boot
A	120	5	0.039	0.058	0.995
B	120	108	0.009	0.060	0.767
C	120	108	0.170	0.063	0.366
D	120	108	0.090	0.066	0.364
E	120	120	0.073	0.066	0.388
A	200	5	0.049	0.054	1.000
B	200	180	0.022	0.060	0.955
C	200	180	0.544	0.050	0.602
D	200	180	0.324	0.069	0.669
E	200	200	0.294	0.076	0.627

Notes: The table reports the main Monte Carlo experiment, based on 1,000 replications and 999 bootstrap draws per replication, seed 20260625. Size rows use $\rho = 1.0$ and power rows use $\rho = 0.5$, read against the restricted wild-bootstrap critical value. “rej @ 5.73” reports rejection against the borrowed Pesaran–Shin–Smith upper bound, the over-rejection the bootstrap corrects, and “rej @ boot” reports rejection against the bootstrap critical value, with target size 0.05 under the null. The replication code also includes a broader grid over alternative sample sizes and nuisance dimensions (function `run_grid`). The results show that rejection against the borrowed 5.73 bound rises sharply in the integrated-nuisance designs C, D, and E (reaching 0.544 at $T = 200$), while the restricted wild-bootstrap rejection rate remains close to the nominal 5 percent level and power rises with T .

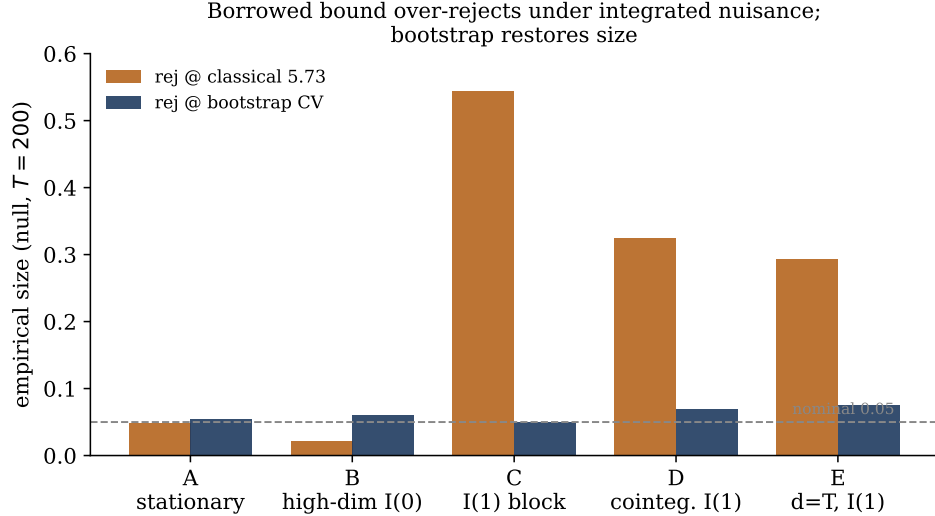


Figure 2: Empirical size at $T = 200$. The classical 5.73 bound over-rejects under the integrated-
nuisance designs, while the restricted wild bootstrap controls size close to the nominal 5 percent
level.

7.3 The feasible test: classical bound versus DML-Bounds bootstrap

Before the oracle-critical-value diagnostics below, we report the comparison that matters in practice: how the *feasible* procedure behaves against the borrowed Pesaran–Shin–Smith bound it is meant to replace. We fix the $d = 40$ nuisance design and, at each sample size, record the rejection rate of the residualized statistic against the classical 5.73 bound and against the restricted wild-bootstrap critical value that DML-Bounds actually uses, under the null ($\rho = 1$, size) and under a moderate alternative ($\rho = 0.5$, power). This is the same statistic, bootstrap, and support re-selection as the main Designs A–E table, applied to the fixed-dimension designs. Table 4 reports the result.

Table 4: Feasible size and power: classical bound versus DML-Bounds bootstrap ($d = 40$)

T	Classical bound (5.73) rejection under H_0	DML-Bounds bootstrap rejection under H_0	Power (bootstrap) under $\rho = 0.5$
<i>Stationary nuisance, pure $I(0)$</i>			
100	0.035	0.072	0.812
250	0.047	0.052	1.000
500	0.060	0.065	1.000
1000	0.047	0.054	1.000
<i>Integrated nuisance, mixed $I(0)/I(1)$ with common trend</i>			
100	0.181	0.061	0.423
250	0.495	0.060	0.982
500	0.658	0.062	0.999
1000	0.737	0.045	1.000

Notes: Rejection rate at the nominal 5% level. Column 2 uses the classical Pesaran–Shin–Smith bound (5.73); column 3 uses the restricted wild-bootstrap critical value. Under stationary nuisance the classical bound is approximately valid; under integrated nuisance it over-rejects, and the bootstrap restores size while column 4 shows power is retained. Entries are computed by the reproducible run (`dml.bounds.robustness.R`, function `table.feasible`).

The integrated panel is the operational claim of the paper made quantitative. Under stationary nuisance the classical bound is approximately valid, with null rejection between 0.035 and 0.060 across sample sizes, and the bootstrap is equally well sized. Under integrated nuisance the borrowed bound over-rejects, and the distortion grows with the sample rather than vanishing: null rejection against 5.73 rises from 0.181 at $T = 100$ to 0.495, 0.658, and 0.737 at $T = 250$, 500, and 1000. At the largest sample the classical bound rejects a true null nearly three-quarters of the time. The feasible bootstrap holds size near nominal throughout the integrated panel, between 0.045 and

0.061, while retaining power that rises from 0.423 at $T = 100$ to unity by $T = 500$. This is the demonstration that the oracle-critical-value tables cannot give: not that the statistic can be sized by an infeasible reference, but that the feasible procedure controls size exactly where the borrowed bound fails, and that the failure is a large-sample phenomenon driven by the integrated controls. The diagnostics that follow hold the inference principle at its infeasible best to isolate the estimator from the bootstrap.

7.4 Endogenous regressors: fixed-regressor versus system bootstrap

The designs above maintain a strongly exogenous focal regressor, so the fixed-regressor and system forms of the bootstrap coincide up to simulation noise. This subsection activates the endogeneity channel that the Pesaran–Shin–Smith conditional model is built for and that Section 6 argues a fixed-regressor wild bootstrap cannot replicate. The design keeps the $d = 40$ mixed $I(0)/I(1)$ nuisance space (half integrated, half stationary AR(1)) and introduces contemporaneous correlation between the marginal innovation of the focal regressor and the equation error: $\Delta D_t = v_t$ with $v_t = \delta e_t + \sqrt{1 - \delta^2} \xi_t$, where e_t is the innovation of the long-run error and ξ_t is independent noise, so $\delta = \text{corr}(v_t, e_t)$ indexes endogeneity, with $\delta = 0$ recovering the exogenous designs above. Both bootstrap schemes are computed on the same simulated data sets and the same statistic, so the columns of Table 5 differ only in the null law each scheme simulates.

Three findings organize the table. First, the sanity rows: at $\delta = 0$ the two schemes agree, both holding empirical size near the nominal level, which confirms that the system regeneration introduces no distortion of its own and that the exogenous results of Tables 3–4 are unaffected by the change of algorithm. Second, both schemes control size across the endogeneity grid, while the borrowed 5.73 bound over-rejects at rates between 0.29 and 0.54 in every null cell. That the fixed-regressor scheme does not visibly deteriorate even at $\delta = 0.8$ has a specific explanation: the restricted null dynamics from which $\hat{e}_t$ is drawn include the contemporaneous ΔD_t term with its estimated coefficient, so the resampled conditional error has already been orthogonalized against the regressor innovation, and the regenerated ΔY^* inherits the correlation through that coefficient rather than through the resampling. This protection, however, is a property of the conditional specification, not of the resampling scheme, and it is exactly the kind of knife-edge adequacy that

Table 5: Endogenous focal regressor: fixed-regressor versus system restricted wild bootstrap ($d = 40$ mixed $I(0)/I(1)$ nuisance)

			rejection rate at nominal 5%		
T	δ	ρ	@ classical 5.73	@ fixed-regressor	@ system
<i>Size ($\rho = 1$)</i>					
120	0.0	1.0	0.333	0.080	0.080
120	0.4	1.0	0.300	0.093	0.093
120	0.8	1.0	0.310	0.073	0.083
200	0.0	1.0	0.533	0.070	0.060
200	0.4	1.0	0.540	0.057	0.063
200	0.8	1.0	0.443	0.067	0.057
<i>Power ($\rho = 0.5$)</i>					
120	0.8	0.5	0.787	0.357	0.350
200	0.4	0.5	1.000	0.963	0.973
200	0.8	0.5	1.000	0.927	0.953
<i>Common trend between D and the integrated controls ($\delta = 0.8$)</i>					
200	0.8	1.0	0.293	0.033	0.027
200	0.8	0.5	0.990	0.783	0.790

Notes: $\delta = \text{corr}(v_t, e_t)$ is the contemporaneous correlation between the marginal innovation of the focal regressor and the long-run equation error; $\delta = 0$ recovers the exogenous designs of Table 3. Both bootstrap schemes are evaluated on the same simulated data sets: the fixed-regressor scheme holds D at its realized path and resamples $\hat{e}_t$ alone, the system scheme applies a common Rademacher weight to $(\hat{e}_t, \hat{v}_t)$ and regenerates (Y^*, D^*) jointly per Algorithm 1. Common-trend rows use the severe near-collinearity design in which a latent $I(1)$ trend drives both the integrated controls and the focal regressor. Entries are based on 300 Monte Carlo replications and $B = 199$ bootstrap draws per replication (Monte Carlo standard error for a true size of 0.05 is approximately 0.013); the replication code (`dml_bounds_endogeneity.R`, function `run_endo_grid`) reproduces the table and scales to production settings.

should not be relied on: it presumes the contemporaneous term fully absorbs the dependence, which the estimated single coefficient need not do under richer marginal dynamics, conditional heteroskedasticity of v_t , or misspecified short-run structure. The system scheme carries the empirical joint law of $(\hat{\epsilon}_t, \hat{v}_t)$ into every draw by construction and therefore does not depend on that presumption. Third, the system scheme pays nothing for this robustness: its size is statistically indistinguishable from the fixed-regressor scheme in every cell, and its power is equal or slightly higher, reaching 0.953 against 0.927 at $(T, \delta) = (200, 0.8)$. The common-trend rows repeat the exercise in the severe near-collinearity design in which the integrated controls and the focal regressor load on a latent common trend; because the marginal model for ΔD^* conditions on the differenced controls, the shared trend is transmitted through the fixed control path, and both schemes are, if anything, mildly conservative there while retaining power near 0.79. Endogeneity of the focal regressor is thus handled where the Pesaran–Shin–Smith framework requires it to be handled, inside the simulated null and by construction rather than by reliance on the conditional specification, and the system form of Algorithm 1 is the default everywhere in this paper.

7.5 Robustness: fixed-dimension sensitivity under oracle critical values

The main grid varies the dimension and integration content while holding the inference reference at the feasible bootstrap. This subsection complements it from the opposite direction: it fixes a moderate stationary nuisance dimension and traces size and power across a wider range of sample sizes and penalties, using *method-specific empirical* (oracle) critical values rather than the bootstrap. The purpose is to isolate the finite-sample behavior of the statistic itself, holding the inference principle at its infeasible best, so that any size distortion seen here is a property of the estimator and not of the bootstrap approximation. The two exercises share the same partially linear integrated data-generating family and the same first-stage machinery; they differ only in the critical value used.

We simulate a partially linear integrated system in which the variables of interest are affected by a dense, high-dimensional nuisance matrix. The 40-variable nuisance matrix W_t consists of stationary $I(0)$ autoregressive processes. The structural long-run error u_t follows an AR(1) process $u_t = \rho u_{t-1} + \epsilon_t$, where $\rho = 1.0$ is the null of no cointegration and $\rho \in \{0.85, 0.50, 0.20\}$ represents weak, moderate, and strong cointegration alternatives. For each $T \in \{100, 250, 500, 1000\}$, we

compare three estimators of the residualized statistic, the unpenalized conditional ECM benchmark, a plain (ℓ_1) LASSO, and the adaptive-LASSO DML-Bounds procedure, under three penalty choices ($\lambda_{\min}$, the geometric midpoint, and λ_{1se}). First-stage nuisance functions are estimated with h -block cross-fitting and Post-LASSO refitting.

To separate size from power, we simulate the null model and compute method-specific empirical 95% critical values for each sample size and penalty, so that rejection frequencies under $\rho = 1.0$ are empirical size and those under $\rho < 1$ are size-adjusted power. Consistent with Section 6, evaluating the unpenalized benchmark against the classical [Pesaran et al. \(2001\)](#) upper bound produces severe size distortion from degrees-of-freedom exhaustion, which is why these tables use method-specific empirical critical values.

Table 6: Empirical Size ($\rho = 1.0$) Under High-Dimensional Confounding ($d = 40$)

T	Penalty	Unpenalized ECM	Plain LASSO	Adaptive (DML-Bounds)
100	Low	0.035	0.065	0.040
250	Low	0.055	0.050	0.050
500	Low	0.075	0.060	0.045
1000	Low	0.035	0.060	0.055
100	Medium	0.035	0.075	0.045
250	Medium	0.055	0.035	0.050
500	Medium	0.075	0.055	0.040
1000	Medium	0.035	0.075	0.060
100	High	0.035	0.040	0.030
250	High	0.055	0.050	0.060
500	High	0.075	0.045	0.050
1000	High	0.035	0.060	0.055

Notes: Rejection rates use method-specific empirical 95% critical values from the null distribution, reporting empirical size rather than size against classical asymptotic bounds. All three methods and rejection rates are computed by the reproducible robustness run (`dml_bounds_robustness.R`).

Table 7: Size-Adjusted Power Under Moderate Cointegration ($\rho = 0.50$)

T	Penalty	Unpenalized ECM	Plain LASSO	Adaptive (DML-Bounds)
100	Low	0.360	0.565	0.480
250	Low	1.000	0.955	0.965
500	Low	1.000	0.995	0.995
1000	Low	1.000	1.000	1.000
100	Medium	0.360	0.470	0.465
250	Medium	1.000	0.935	0.930
500	Medium	1.000	0.995	0.995
1000	Medium	1.000	1.000	0.995
100	High	0.360	0.320	0.265
250	High	1.000	0.815	0.835
500	High	1.000	0.935	0.945
1000	High	1.000	0.970	0.990

Notes: Power is size-adjusted using method-specific empirical 95% critical values. The unpenalized benchmark’s good performance here relies on infeasible empirical critical values.

The DML-Bounds procedure remains close to the nominal level and shows power approaching one for moderate and large samples. The favorable size-adjusted power of the unpenalized benchmark should be read with caution: size adjustment re-centers a poorly behaved finite-sample statistic using an infeasible empirical critical value, whereas the practical problem is that classical bounds tables are applied directly. The operational advantage of DML-Bounds is feasibility and stability.

7.6 Robustness to non-stationary confounding

To assess robustness, we consider a mixed $I(0)/I(1)$ design in which half of the 40 confounders are non-stationary random walks and a latent common trend drives both the non-stationary confounders and the core regressor, producing severe near-collinearity. This is the mixed cell of the integration axis and the regime in which the trend-absorption bracket of Section 3 is most active. Here the empirical critical values required to control size for the unpenalized benchmark become extremely large, while DML-Bounds remains close to nominal and recovers power as the sample grows.

Table 8: Empirical Size ($\rho = 1.0$) Under Mixed $I(0)/I(1)$ Confounding ($d = 40$)

T	Penalty	Unpenalized ECM	Plain LASSO	Adaptive (DML-Bounds)
100	Low	0.030	0.075	0.040
250	Low	0.035	0.075	0.050
500	Low	0.025	0.045	0.040
1000	Low	0.035	0.045	0.040
100	Medium	0.030	0.055	0.060
250	Medium	0.035	0.060	0.040
500	Medium	0.025	0.030	0.060
1000	Medium	0.035	0.060	0.075
100	High	0.030	0.085	0.010
250	High	0.035	0.040	0.070
500	High	0.025	0.070	0.040
1000	High	0.035	0.055	0.065

Notes: Method-specific empirical 95% critical values; empirical size rather than size against classical bounds. All three methods and rejection rates are computed by the reproducible robustness run (`dml_bounds_robustness.R`).

Table 9: Size-Adjusted Power ($\rho = 0.50$) Under Mixed $I(0)/I(1)$ Confounding

T	Penalty	Unpenalized ECM	Plain LASSO	Adaptive (DML-Bounds)
100	Low	0.100	0.185	0.110
250	Low	0.980	0.945	0.920
500	Low	1.000	1.000	1.000
1000	Low	1.000	1.000	1.000
100	Medium	0.100	0.185	0.175
250	Medium	0.980	0.925	0.920
500	Medium	1.000	1.000	1.000
1000	Medium	1.000	1.000	1.000
100	High	0.100	0.125	0.110
250	High	0.980	0.830	0.770
500	High	1.000	1.000	1.000
1000	High	1.000	1.000	1.000

Notes: Power is size-adjusted using method-specific empirical 95% critical values. The unpenalized benchmark performs well only after recentring with infeasible empirical critical values. All three methods and rejection rates are computed by the reproducible robustness run (`dml_bounds_robustness.R`).

The critical values themselves, not only the rejection rates, carry the message. Under stationary nuisance the empirical 95% critical value of the DML-Bounds statistic stays close to the classical Pesaran–Shin–Smith bound, rising from about 2.4 at $T = 100$ toward 5.4 as T grows, against a

borrowed bound near 5.73. Under the mixed integrated design it rises far above that bound and increases with the sample, from about 5.1 at $T = 100$ to 19.4 at $T = 1000$, and the unpenalized benchmark requires the largest values of all, 14.4 at $T = 100$ where the adaptive procedure requires 5.1, rising to 20.8 at $T = 1000$ (medium penalty). This is the finite-sample counterpart of the inference problem: once integrated controls are partialled out, the realized null is far more dispersed than the classical table, so a borrowed critical value over-rejects and a method-specific or bootstrap critical value is required. The unpenalized benchmark attains competitive power in the tables above only because it is recentred with an infeasibly large empirical critical value that is unavailable in practice. (The full set of empirical critical values is reported in the `cv95` column of the saved `res$sizepower` object.)

7.7 Ultra-high-dimensional implementability check

To push the comparison further, we consider an ultra-high-dimensional regime with $T = 100$ and $d = 150$, where the nuisance space mixes stationary and integrated variables but the active set is sparse. Here the empirical Gram matrix of the unpenalized benchmark is singular and the classical conditional ECM cannot be implemented, while the regularized DML-Bounds procedure remains estimable.

Table 10: Ultra-High-Dimensional Implementability Check ($T = 100, d = 150$)

Method	Implementable across draws	Statistic (median, IQR)
Unpenalized ECM	0%	not defined
DML-Bounds (h-block)	97.5%	0.76 (1.24)

Notes: At $T = 100, d = 150$ the unpenalized conditional ECM is rank-deficient (singular Gram matrix) in every replication, so no statistic is defined and it is never implementable. The regularized DML-Bounds procedure remains estimable in essentially all replications and delivers a non-degenerate statistic; the table reports the share of replications in which each method is implementable and, for DML-Bounds, the median and interquartile range of the statistic across the implementable draws. The purpose is implementability, not a size or power comparison, so the statistic is summarized by its distribution rather than a single run. Entries are computed by the reproducible robustness run (`dml.bounds_robustness.R`, function `table.ultra`).

This is an implementability check rather than a complete finite-sample characterization. It shows only that the regularized statistic can be computed where the unpenalized ECM cannot; it is not evidence on size or power when $d > T$. Overall, the simulations support the central

interpretation: the objective is not to replace the classical ARDL bounds test, which remains well suited to low-dimensional systems, but to provide a feasible implementation of the same projection logic, with bootstrap inference, when the nuisance space becomes too large or too collinear for unpenalized estimation.

8 Application: Exchange-Rate Pass-Through

Exchange-rate pass-through to prices is a data-rich cointegration problem, and the natural main application for a method built for high-dimensional persistent conditioning sets. A canonical consumption–income illustration, a more familiar but less demanding benchmark, is reported in Appendix C. Here the long-run relation between the price level and the exchange rate must be assessed conditional on a large set of macroeconomic and financial controls, many of which are themselves persistent or integrated. This is exactly the environment in which the trend-absorption bracket is informative: some controls may share stochastic trends with the tested variables, so the effective integrated count $\tilde{k}$, and hence the relevant null distribution, is not known in advance.

Concretely, the application takes the log U.S. consumer price index as the dependent variable Y_t and the log U.S. nominal effective exchange rate as the focal regressor D_t , both monthly. The conditioning set W_t comprises seven macroeconomic and financial controls: the broad money stock (M2), the federal funds rate, industrial production, the unemployment rate, the West Texas Intermediate oil price, the ten-year Treasury yield, and the Baa corporate yield. Several of these are persistent or integrated and may share stochastic trends with the dollar. The procedure follows the methodology of Section 4: the balanced first stage residualizes price differences on stationary controls and the lagged levels on control levels, the adaptive m_Z projection is the default, and inference is the restricted system wild bootstrap of Algorithm 1 with $B = 999$ and lag order $p = 4$. The system form matters here on economic grounds, not only formal ones: exchange-rate innovations are not plausibly exogenous to price shocks, since monetary policy responds to inflation and the dollar responds to monetary policy within the month, so the marginal innovations of the exchange rate and the conditional error of the price equation are exactly the pair whose correlation the joint regeneration preserves. We run the test over four monetary-policy regimes, so that any relation is assessed within a regime rather than across structural breaks, and we report the trend-absorption

Table 11: Exchange-rate pass-through to U.S. prices across monetary regimes. The focal relation is between the log price level and the log nominal effective exchange rate. “Full” conditions on all seven controls; “reduced” drops money and the oil price. Critical values are the 95th percentile of the restricted wild bootstrap; p is the bootstrap p -value.

Regime	Conditioning set	n	F	Bootstrap cv _{.95}	p	Verdict
1973–1985	full (7)	156	1.51	10.64	0.79	fail to reject
	reduced (5)	156	0.45	5.23	0.84	fail to reject
1986–1998	full (7)	156	1.03	3.75	0.49	fail to reject
	reduced (5)	156	0.38	3.94	0.76	fail to reject
1999–2007	full (7)	108	4.70	8.07	0.31	fail to reject
	reduced (5)	108	5.70	4.96	0.03	reject
2008–2020	full (7)	156	3.86	7.71	0.32	fail to reject
	reduced (5)	156	1.43	5.03	0.50	fail to reject

diagnostic of Definition 2 by contrasting the full conditioning set against a reduced set that drops the two controls most likely to be cointegrated with the pass-through relation, namely money and the oil price.

Table 11 reports the results. In three of the four regimes the test fails to reject under both conditioning sets, which is consistent with weak long-run pass-through to U.S. prices outside of specific episodes. The informative regime is the late Great Moderation, 1999–2007. There the full seven-control set produces a high bootstrap critical value, 8.07, that reflects a large effective integrated count $\tilde{k}$, so the observed $F = 4.70$ falls well short and the test fails to reject. Dropping money and the oil price lowers $\tilde{k}$, the bootstrap critical value falls to 4.96, and the same data now reject at the five percent level, $F = 5.70$ with $p = 0.03$, recovering a long-run pass-through elasticity of 0.29 with a standard error of 0.13. The verdict flips not because the data change but because the reduced set no longer absorbs the trend the focal relation lives on. The pattern is what the over-absorption mechanism of Section 5 predicts, though as with any change of conditioning set it cannot by itself rule out ordinary specification sensitivity; the theory supplies the interpretation, the contrast supplies the evidence consistent with it. Against the borrowed PSS bound the full-set statistic would be read against the wrong null, and the trend-absorption bracket is what separates the two readings.

This setting sharpens the method for a specific reason. In the canonical consumption–income benchmark of Appendix C the natural controls are broad macro aggregates that plausibly span the tested relation, so the redundancy caveat of Assumption 5 binds and the verdict is control-

sensitive throughout. In the pass-through system the focal relation is between the price level and the exchange rate, while the conditioning set is financial and external-sector variables that are mostly distinct from that relation, so the diagnostic gap is small in three regimes and becomes material only in the Great Moderation, where money and commodity prices most plausibly shared trends with the dollar. The diagnostic thus does what it is meant to do: it stays quiet when the controls are benign and flags the one regime where conditioning changes the answer.

9 Conclusion

This paper develops a bounds-testing theory for cointegration after high-dimensional orthogonalization. The organizing idea is the effective integrated count $\tilde{k}$: residualizing lagged levels against a nuisance space that may itself be integrated can absorb stochastic trends, so the quantity governing the test is the number of trends that survive residualization rather than the integration order of the original regressors. Stationary controls leave the upper-bound logic intact; integrated controls can move the statistic inside the bounds interval. The same Pesaran–Shin–Smith interval that brackets the integration order of the regressors reappears as a bracket over surviving stochastic trends.

The theory is built from the scalar error-correction root upward, so that each high-level condition is the limit of a case that can be checked by hand, and is organized as a ladder. An oracle theorem identifies the limit law, indexed by $\tilde{k}$; a feasibility theorem shows the cross-fitted statistic matches the oracle to $o_p(1)$ under sparse linear nuisance structure, a balanced first-stage specification, split compatibility, and cross-fit decoupling; and a corollary recovers the classical bounds test as the corner in which the nuisance space carries no integrated component. Because the equivalence is first-order and the remainder is non-negligible at empirically relevant sample sizes, the classical bounds tables are useful conceptually but not operationally valid; valid inference is obtained through a restricted system wild bootstrap that regenerates the dependent variable and the focal regressor jointly under the null, applying a common wild weight to the conditional and marginal residuals so that the endogeneity correlation of the Pesaran–Shin–Smith setting is preserved in the simulated null, recomputes the full residualized statistic, and conditions on the realized trend content of the nuisance space. The bootstrap is presented as Algorithm 1, a restricted system wild-bootstrap procedure in the spirit of [Cavaliere et al. \(2012\)](#), motivated by the oracle and equivalence results,

proved consistent in the oracle fixed-integrated-block design for arbitrary endogeneity correlation (Theorem 2), and evaluated directly in simulations, including designs with strong contemporaneous correlation between the regressor innovations and the equation error; the remaining step, uniformity of bootstrap validity over general first-stage learners and mixed integrated nuisance spaces, is left to future work.

The results are intentionally scoped. The paper establishes oracle-equivalence under sparse balanced nuisance structure and evaluates the restricted bootstrap through simulations designed to match the theoretical mechanisms; in the empirically relevant cases considered here, recomputing the full residualized statistic under the null delivers substantially better size control than the classical bounds critical values. The boundaries of the contribution are stated once and plainly. The functional limit on the surviving trend that identifies $\tilde{k}$ is verified by hand in the scalar and bivariate cases rather than for arbitrary mixed nuisance spaces; the no-redundancy condition of Assumption 5 cannot be checked directly, which is why the paper supplies the trend-absorption diagnostic of Definition 2 as a practical substitute; and the growth condition $s \log d / \sqrt{T} \rightarrow 0$ is the price of combining nonstationary residualization with cross-fit decoupling. A fully general bootstrap invariance theorem for arbitrary learners and mixed integrated nuisance spaces is left to future work. On the empirical side, the long-run coefficient is more sensitive to regularization than the testing decision, and the canonical consumption–income verdict of Appendix C is lag- and control-sensitive, which the pass-through application sharpens by isolating a single regime in which the diagnostic changes the verdict. That sensitivity is itself the point: valid bootstrap inference removes a high-dimensional statistic a classical reading would call decisive, corrects false rejections of the borrowed bound, and, through the full-versus-reduced control contrast, shows the long-run relation masked when trend-sharing aggregates are included and revealed when they are dropped. The method’s role is to make that structure legible rather than to resolve it by fiat. Even so, the results suggest that high-dimensional orthogonalization, read through the trend-absorption bracket and implemented with bootstrap inference, offers a promising route for extending classical cointegration testing to modern data-rich environments.

A Technical derivations

This appendix records the key step of each result; full intermediate algebra follows the cited standard references and is omitted to keep the exposition compact.

A.1 Lemma 1 (stationary residualization preserves the trend). Let Z_{t-1} be $I(1)$ and let it be residualized on a stationary regressor set with fitted coefficient $\hat{c}$. Because the regressors are stationary, $\hat{c} = O_p(1)$ and the fitted component w_t satisfies $\max_{t \leq T} |w_t| = o_p(\sqrt{T})$. Hence

$$\tilde{Z}_{t-1} = Z_{t-1} - w_t = Z_{t-1} + o_p(\sqrt{T}),$$

so the residualized regressor retains the original $I(1)$ trend at the $\sqrt{T}$ scale: stationary partialling cannot remove a stochastic trend.

A.2 Proposition 1 (effective integrated count). Let the k tested integrated regressors Z be residualized against an integrated nuisance block W_1 whose span shares r independent cointegrating directions with Z , $0 \leq r \leq k$. Projection removes exactly those r common stochastic trends, so the residualized $\tilde{Z}$ has

$$\tilde{k} = k - r$$

surviving integrated directions. The two extremes $r = 0$ (stationary nuisance, no absorption) and $r = k$ (full absorption) are the endpoints of the trend-absorption bracket; the classical Pesaran–Shin–Smith case is $r = 0$.

A.3 Proposition 3 (oracle residualized limit). Writing $\dot{Z}_{t-1}$ for the surviving $\tilde{k}$ -dimensional integrated part of $\tilde{Z}_{t-1}$, standard functional central limit and continuous-mapping arguments give

$$T^{-2} \sum_t \dot{Z}_{t-1} \dot{Z}_{t-1}' \Rightarrow \int_0^1 B(r) B(r)' dr, \quad T^{-1} \sum_t \dot{Z}_{t-1} \varepsilon_t \Rightarrow \int_0^1 B(r) dM(r),$$

for a $\tilde{k}$ -vector Brownian motion B and martingale limit M . The oracle bounds statistic is therefore the classical bounds functional indexed by $\tilde{k}$ rather than by k .

A.4 Proposition 4 (feasible equals oracle). The only object separating the feasible cross-fitted statistic from the oracle is the cross term between the residualized levels and the first-stage error r_t . Under the balanced specification, cross-fit decoupling (Lemma 2) and the sparse Post-LASSO rate give

$$\frac{1}{T} \sum_t \tilde{Z}_{t-1} r_t = O_p\left(\frac{s \log d}{\sqrt{T}}\right) = o_p(1)$$

under the growth condition $s \log d / \sqrt{T} \rightarrow 0$, so the feasible statistic equals the oracle statistic up to $o_p(1)$ and inherits the limit of Proposition 3.

A.5 Theorem 2 (bootstrap validity, oracle fixed-integrated-block case). The argument has three steps, each an application of established wild-bootstrap theory for nonstationary data (Cavaliere and Taylor, 2009; Cavaliere et al., 2012; Smeekes, 2013).

Step 1: conditional invariance principle for the stacked innovations. Conditionally on the data, $(\epsilon_t^*, v_t^*) = \eta_t(\hat{\epsilon}_t, \hat{v}_t)$ is a martingale difference array with respect to the bootstrap filtration, and because $\eta_t^2 = 1$ the conditional second-moment matrix is

$$\frac{1}{T} \sum_t \begin{pmatrix} \hat{\epsilon}_t^2 & \hat{\epsilon}_t \hat{v}_t \\ \hat{\epsilon}_t \hat{v}_t & \hat{v}_t^2 \end{pmatrix} \rightarrow_p \Sigma,$$

by consistency of the restricted estimates under H_0 (condition (ii)) and the law of large numbers for the squares and cross products, which the finite fourth moments of condition (i) deliver. This is the step at which the common wild weight matters: applying independent weights to the two residual series would drive the off-diagonal conditional moment to zero regardless of Σ , which is precisely the defect of the fixed-regressor scheme. A conditional functional central limit theorem for wild-bootstrap partial sums then gives, in probability,

$$\frac{1}{\sqrt{T}} \sum_{t \leq \lfloor rT \rfloor} (\epsilon_t^*, v_t^*)' \Rightarrow_p (B_\epsilon^*(r), B_v^*(r))',$$

a bivariate Brownian motion with covariance Σ , the same joint law, including the endogeneity correlation, as the limit of the sample partial sums under H_0 .

Step 2: the bootstrap paths reproduce the null trend configuration. Cumulating v^* plus the

fixed-control exogenous component of the marginal fit makes $T^{-1/2}D_{[rT]}^*$ converge, conditionally in probability, to the same Brownian limit as D under the null; because d_1 is fixed and the projection Π is the oracle one, the common-trend component transmitted through the fixed control path preserves the realized cointegration configuration between the focal regressor and the integrated controls, hence the realized $\tilde{k}$. The regenerated Y^* is built from the restricted null dynamics with consistent coefficients, so its partial sums obey the null invariance principle, and the oracle-residualized bootstrap regressors satisfy the functional central limit theorem on the same $\tilde{k}$ surviving integrated directions as in Assumption 3.

Step 3: continuous mapping and Pólya. The bootstrap statistic F^* is the same continuous functional of the bootstrap quadratic forms as in Proposition 3, so the conditional continuous mapping theorem yields $F^* \Rightarrow_p$ the $\tilde{k}$ -indexed bounds functional, which is the null limit of F by Theorem 1. Weak convergence in probability to a limit with continuous distribution function (condition (iii)) upgrades, by the standard Pólya-type argument, to uniform convergence of the conditional distribution function, which is the displayed sup-norm statement; consistency of the bootstrap critical values and asymptotic uniformity of the bootstrap p -value follow.

B Monte Carlo implementation details

Data-generating process. The core system is $Y_t = D_t + u_t$ with D_t a driftless random walk and $u_t = \rho u_{t-1} + e_t$, where $\rho = 1$ is the no-cointegration null and $\rho = 0.5$ the moderate alternative. The nuisance matrix W_t has a fraction frac_{I1} of $I(1)$ random-walk columns and the remainder stationary AR(1) with coefficient 0.5. In the common-trend designs a latent $I(1)$ trend drives the $I(1)$ nuisance columns and loads on the core regressor, producing the severe near-collinearity case. Designs A–E fix $(d, \text{frac}_{I1}, \text{common trend})$ as in Table 2, with a 100-observation burn-in discarded.

Estimation. The first stage uses h -block cross-fitting with Post-LASSO refitting. The m_Z projection is an adaptive LASSO with marginal (univariate) slope weights and penalty $\lambda = c \sqrt{\log d/n} \hat{\sigma}$, $c = 1.1$; design A uses the low-dimensional OLS projection. Degrees of freedom charge the selected support sizes, and a cell returns a missing value when the residual degrees of freedom are non-positive, which is how the unpenalized $d > T$ case is recorded as not implementable.

Bootstrap. Inference uses the restricted system wild bootstrap of Algorithm 1. Two auxiliary models are estimated: the restricted conditional model for ΔY_t under the imposed null, with residuals $\hat{\epsilon}_t$, and a marginal model for ΔD_t on an intercept, its own lag, and the first-stage-selected differenced controls, with residuals $\hat{v}_t$. On each draw a single Rademacher sequence multiplies the stacked pair $(\hat{\epsilon}_t, \hat{v}_t)$, preserving their empirical contemporaneous covariance and conditional heteroskedasticity; ΔD^* is regenerated from the marginal dynamics with the control path fixed and cumulated to D^* , so the regenerated focal regressor is integrated by construction and inherits any common-trend component with the controls through the fixed control path; ΔY^* is then regenerated from the restricted conditional dynamics driven by ΔD^* and ϵ^* , cumulated to Y^* , and the full residualized statistic is recomputed on (Y^*, D^*, W) with W held fixed. The stationary first-stage support is frozen, while the Y -side and D -side level supports are *re-selected on each bootstrap path*, so that selection error in the integrated directions is reflected in the bootstrap law; freezing them instead leaves a size distortion that grows with T . The endogeneity designs of Section 7.4 set $v_t = \delta e_t + \sqrt{1 - \delta^2} \xi_t$ in the data-generating process, so that δ is the contemporaneous correlation between the marginal innovation of D and the equation error; the fixed-regressor comparison scheme differs from the system scheme only in holding D at its realized path and resampling $\hat{\epsilon}_t$ alone. The main-grid designs of Tables 3–4 maintain a strongly exogenous focal regressor, under which the two schemes coincide; those tables are computed under the fixed-regressor implementation, which Algorithm 1 nests as the strong-exogeneity special case, and the $\delta = 0$ rows of Table 5 verify the equivalence numerically. The bootstrap critical value is the 95th percentile of the finite $\{F_b^*\}$.

Settings. The main experiment uses $R = 1,000$ Monte Carlo replications, $B = 999$ bootstrap draws, sample sizes $T \in \{120, 200\}$, and seed 20260625. The replication code also provides a broader grid (`run_grid`) over alternative sample sizes and nuisance dimensions, and a reproducible parallel driver that assigns one L’Ecuyer-CMRG stream per cell so results are identical across core counts.

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C Canonical consumption–income illustration

This appendix reports a canonical sanity check on a familiar benchmark, a deliberately undemanding setting in which the conditioning set does not strongly stress the method. We revisit the relationship between U.S. real consumption, $Y_t = \log(\text{PCECC96})$, and real personal income, $D_t = \log(\text{RPI})$. Identifying long-run macroeconomic equilibria often requires conditioning on a rich set of short-run dynamics and macroeconomic controls, so we augment the conditional error-correction specification with a high-dimensional nuisance vector containing four lags of ten macroeconomic variables.

C.1 Baseline result and the role of inference

Table 12 reports the headline comparison at the preferred specification (four lags, medium penalty). The classical unpenalized ECM yields an F -statistic of 1.383, which does not reject. A naive high-dimensional implementation that residualizes every variable on stationary differences yields a much

larger statistic, but, as Section 3 shows, such a specification cannot absorb stochastic trends and inflates the statistic through spurious regression; it is not the balanced procedure of this paper. The correct balanced DML-Bounds statistic, which residualizes the lagged levels against control levels, is far smaller, and once it is read against its bootstrap null it does not reject.

Table 12: Empirical Results: U.S. Consumption vs. Income (four lags, medium penalty)

Method	Long-Run MPC ($\hat{\theta}$)	Cointegration Statistic	Decision
ARDL (Unpenalized)	0.829	1.383 (F)	Fail to reject
DML-Bounds, OLS m_Z	0.450	2.179 (F)	Fail to reject
DML-Bounds, adaptive m_Z	0.801	1.097 (F)	Fail to reject

Notes: Decisions are bootstrap decisions (Table 13), not comparisons with the classical Pesaran et al. (2001) 5% Case III bounds ($I(0) = 4.94$, $I(1) = 5.73$), which Section 6 shows are not operationally valid here. The adaptive m_Z row is the headline; the long-run MPC of 0.801 (delta-method SE 0.181) is economically sensible, whereas the OLS- m_Z MPC is less stable.

At the preferred specification, the long-run consumption-income relation is not detected once inference is valid: the bootstrap p -values are 0.76 (adaptive) and 0.58 (OLS). This is not the whole story, since the verdict turns out to depend on both lag length and the control set, as Section C.2 shows. What is unambiguous here is the contrast with the naive reading: comparing the high-dimensional statistic to 5.73 would have declared cointegration, whereas the balanced statistic read against its own null does not. Table 13 reports the restricted-wild-bootstrap critical values and p -values under both m_Z projections.

Table 13: Bootstrap Inference for the U.S. Consumption–Income Relationship

m_Z projection	Observed F	Bootstrap CV (95%)	Bootstrap p -value	Decision
Unpenalized (OLS) m_Z	2.179	6.162	0.577	Fail to reject
Adaptive LASSO m_Z	1.097	5.738	0.760	Fail to reject

Notes: Four lags, medium penalty, full control set. Critical values and p -values from $B = 999$ restricted wild bootstrap replications (Section 6). Both bootstrap critical values exceed the classical 5.73 bound, and the verdict is the bootstrap decision, not the comparison with 5.73.

C.2 Sensitivity, false rejections, and the redundancy diagnostic

Table 14 reports the full-control grid across lag lengths $p \in \{4, 6, 8, 10\}$, penalties, and both m_Z selectors, with the bootstrap critical value and p -value alongside each statistic. Three patterns stand out, and a fourth emerges from the control-set robustness check below.

First, the bootstrap critical values range from 5.7 to 22.5, almost all above the classical 5.73, which is the trend-absorption bracket made empirical: the finite-sample null sits well above the borrowed bound, by a margin that grows with the lag length and the dimension of the nuisance.

Second, the central thesis of the paper is visible directly. In ten of the twelve specifications where the statistic exceeds 5.73, a classical reading would declare cointegration, yet the bootstrap fails to reject. The clearest are $p = 10$ high under both selectors ($F = 12.5$ and 14.3 against bootstrap critical values of 17.1 and 22.5 , $p = 0.20$ and 0.27) and $p = 6$ high OLS ($F = 9.1$ against 9.2 , $p = 0.05$). These are false rejections of the classical bound that the bootstrap corrects.

Third, with the full control set, genuine rejection survives in only one cell, $p = 8$ medium ($p = 0.041$ adaptive, 0.034 OLS). On its own this reads as weak, lag-sensitive evidence. The long-run MPC under the adaptive selector is economically sensible and stable across the grid (0.76 to 0.89), while under OLS m_Z it is more volatile, the diagnostic favoring the adaptive projection.

Table 14: Sensitivity and Bootstrap Inference: U.S. Consumption vs. Income (full controls)

p	Penalty	m_Z	F	Boot CV (95%)	Boot p	MPC (SE)
4	low	adaptive	5.887	7.733	0.158	0.875 (0.087)
4	low	OLS	4.519	7.928	0.287	0.662 (0.172)
4	medium	adaptive	1.097	5.738	0.760	0.801 (0.181)
4	medium	OLS	2.179	6.162	0.577	0.450 (0.354)
4	high	adaptive	2.486	9.636	0.829	0.888 (0.157)
4	high	OLS	4.453	9.040	0.480	0.667 (0.172)
6	low	adaptive	6.004	7.969	0.134	0.825 (0.081)
6	low	OLS	5.653	7.768	0.162	0.613 (0.165)
6	medium	adaptive	6.045	9.278	0.234	0.855 (0.085)
6	medium	OLS	4.941	9.384	0.347	0.606 (0.179)
6	high	adaptive	2.789	10.116	0.783	0.783 (0.135)
6	high	OLS	9.147	9.209	0.052	0.717 (0.117)
8	low	adaptive	5.572	10.074	0.299	0.785 (0.085)
8	low	OLS	6.205	8.819	0.188	0.530 (0.180)
8	medium	adaptive	10.721	10.162	0.041	0.764 (0.061)
8	medium	OLS	10.399	9.881	0.034	0.654 (0.118)
8	high	adaptive	2.524	9.924	0.807	0.773 (0.146)
8	high	OLS	6.205	8.644	0.165	0.530 (0.180)
10	low	adaptive	5.413	8.057	0.254	0.774 (0.089)
10	low	OLS	6.745	9.017	0.157	0.384 (0.221)
10	medium	adaptive	5.626	10.167	0.298	0.764 (0.087)
10	medium	OLS	7.081	8.686	0.143	0.453 (0.191)
10	high	adaptive	12.467	17.091	0.200	0.853 (0.063)
10	high	OLS	14.317	22.490	0.265	0.951 (0.111)

Notes: $B = 999$ restricted wild bootstrap with the Y-side level support re-selected on each path. “Boot p ” is the bootstrap p -value; bold marks rejection at 5%. MPC is the long-run coefficient with delta-method standard error. Rows where $F > 5.73$ but “Boot p ” > 0.05 are classical false rejections corrected by the bootstrap.

The fourth pattern is the most informative, and it comes from re-running the entire grid on a reduced control set that drops the two nominal aggregates, $M2$ and the consumer price index, which are the controls most likely to share a stochastic trend with the consumption-income relation itself. Table 15 contrasts the two control sets at two representative cells. At short lags ($p = 4$, medium) the fail-to-reject is robust: dropping the nominal aggregates leaves the verdict unchanged ($p = 0.76$ versus 0.69) and in fact tightens the long-run MPC from 0.80 to 0.90. At long lags ($p = 10$, medium) the verdict reverses: with the full set the test fails to reject ($F = 5.6$, $p = 0.30$), while dropping $M2$ and CPI it rejects decisively ($F = 15.7$, $p = 0.012$ adaptive; $F = 21.2$, $p = 0.001$ OLS), with a tight and sensible MPC near 0.9.

Table 15: Control-Set Robustness: Full vs. Reduced Nuisance (medium penalty)

p	Controls	m_Z	F	Boot CV (95%)	Boot p	MPC (SE)
4	Full (8)	adaptive	1.097	5.738	0.760	0.801 (0.181)
4	Drop $M2$, CPI (6)	adaptive	1.242	5.996	0.686	0.904 (0.125)
10	Full (8)	adaptive	5.626	10.167	0.298	0.764 (0.087)
10	Drop $M2$, CPI (6)	adaptive	15.741	12.307	0.012	0.925 (0.036)
10	Drop $M2$, CPI (6)	OLS	21.229	13.081	0.001	0.708 (0.056)

Notes: $B = 999$ restricted wild bootstrap. Bold marks rejection at 5%. The short-lag verdict is robust to dropping the nominal aggregates; the long-lag verdict reverses, a pattern consistent with the redundancy mechanism of Assumption 5.

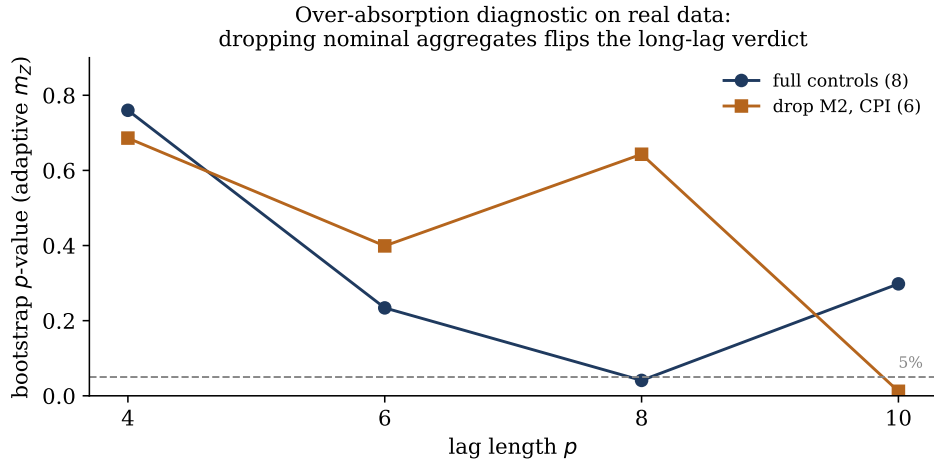


Figure 3: Trend-absorption diagnostic on the U.S. consumption–income data (adaptive m_Z , medium penalty). At short lags the bootstrap p -value is robust to dropping the nominal aggregates; at $p = 10$ the verdict reverses, with the reduced set rejecting, a pattern consistent with over-absorption under Assumption 5.

Read together, the evidence is genuinely lag- and control-sensitive, and the honest summary is that the method makes that sensitivity visible rather than resolving it. The naive high-dimensional statistic, far above any tabulated bound, is an artifact that never returns under valid inference; the classical 5.73 bound would falsely reject in most cells where the statistic exceeds it; at short lags no long-run relation is detected under any control set; and at long lags a stable relation with an MPC near 0.9 emerges precisely when the trend-sharing nominal aggregates are removed. The last fact is the redundancy caveat of Assumption 5 in its empirically observable form, a pattern

consistent with, though not by itself proof of, trend absorption: a control that is cointegrated with the tested relation absorbs it, so including $M2$ and CPI masks at long lags what the reduced set reveals. The gap between the full and reduced control sets, like the gap between the adaptive and OLS m_Z projections, is therefore not noise but a usable diagnostic for over-absorption. A cleaner test of the method, in which the conditioning set is more clearly distinct from the tested relation, motivates the main application of Section 8.